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# 5 Canine Anatomy

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This text is intended for people who already possess knowledge of either veterinary or human anatomy. To assist communication among human rehabilitation and veterinary colleagues, some anatomic terms used for dogs appear in regular print with the analogous terminology for humans in parentheses following the canine term. These comparisons have been minimized, as this is a chapter about canine anatomy and not a chapter about comparative anatomy. Comparative anatomy between dogs and humans has been described in other sources.[1-3]

We have chosen to use some terms consistently throughout the chapter, rather than use equally acceptable synonyms. The canine forelimb is known also as the *thoracic limb* and the *pectoral limb,* but we use the term *forelimb.* The canine hindlimb is known also as the *pelvic limb* or *rear limb,* but we use the term *hindlimb.* Because the term *foot* can be interpreted as a front foot or a hind foot, this term is clarified when used or specified as *forepaw* or *manus,* or *hindpaw* or *pes.* The terms *trunk, neck,* and *head* refer to the same body segments in dogs and humans. The word *canine* is an adjective and the word *dog* is a noun; these terms are used in this consistent grammatical form throughout the chapter.

## Directional Terms and Anatomic Planes

### Directional Terms from Normal Stance (Anatomic Position)

The dog stands upright on digits or phalanges of each forepaw or manus and each hindpaw or pes (Figure 5-1). This type of stance is termed a *digitigrade stance.* The human stands upright on the feet, with the plantar aspect of the feet contacting the floor and adjacent to each other. The upper limbs hang at the sides of the body, palms facing forward. This type of stance is called a *plantigrade stance.*

Directional terms from anatomic position in dogs are more directly compared with the directional terms in humans when the human is in a quadruped position or the dog is in an upright stance posture. Directional terms include *cranial, caudal, rostral, dorsal, palmar, plantar, medial,* and *lateral.* Other specific directional terms include (1) radial and ulnar to indicate toward the radius and ulna, respectively; (2) axial and abaxial to indicate toward or away from the axis of the digits, which is between the third and fourth digits of the forepaw, and the third and fourth digits of the hind paw, respectively; and (3) tibial and fibular to indicate toward the tibia and fibula, respectively.

### Anatomic Planes

The main planes of motion for dogs are as follows (see Figure 5-1):

- The *sagittal plane* divides the dog into right and left portions. If this plane were in the midline of the body, this is the median plane or median sagittal plane.
- The *dorsal plane* divides the dog into ventral and dorsal portions.
- The *transverse plane* divides the body into cranial and caudal portions.

### Axes of Rotation

Motion may occur in any of three planes of motion or some combination. Joint motion within a plane usually occurs around an axis of rotation, which may be centered within the joint space or within the bone comprising the joint. Some joint motions are planar or gliding motions and do not occur around an axis of rotation.

An axis of rotation for a joint motion is a straight line or rod that is 90 degrees to the plane of motion. For each axis of rotation listed in the next section, the plane of motion around which joint motion occurs can be viewed from Figure 5-1.

#### *Axes of Rotational Joint Motion*

The axes of rotational joint motion are as follows:

- *Transverse axis:* Sagittal plane motion occurs around an axis of rotation that is directed mediolaterally.
- *Ventrodorsal axis:* Dorsal plane motion occurs around an axis of rotation that is directed ventrodorsally.
- *Craniocaudal axis:* Transverse plane motion, such as rotation of the trunk, occurs around an axis of rotation that is directed craniocaudally.

Most joints allow motion in more than one plane. Dogs and humans have the ability to selectively produce motion in one, some, or all of the planes of motion at one time.

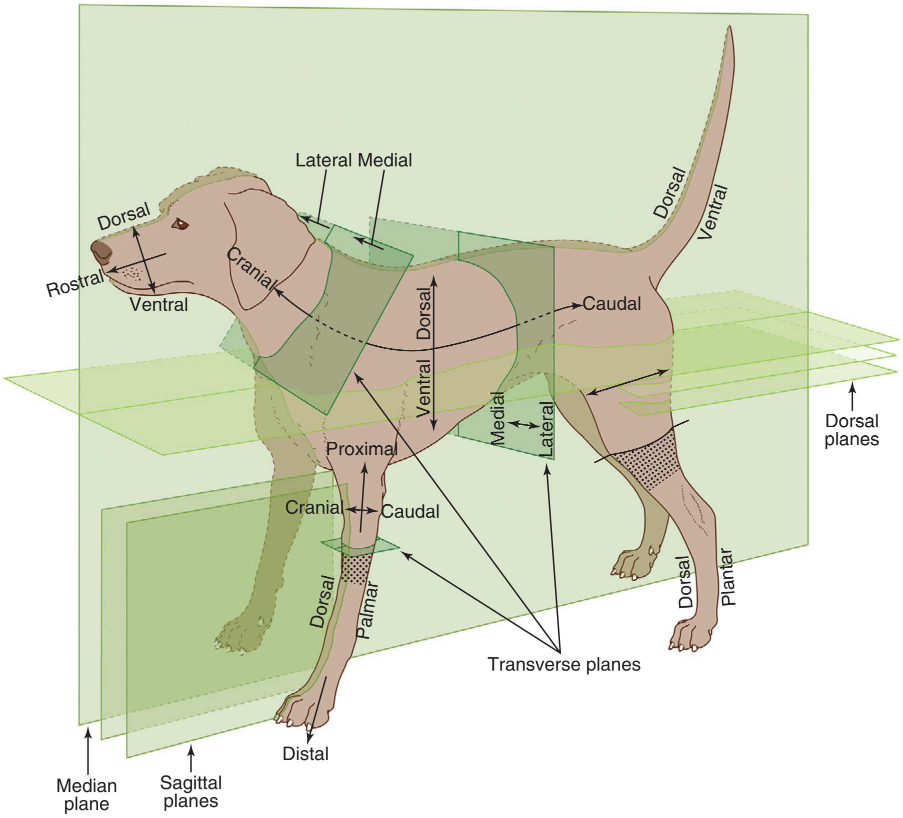


**Figure 5-1** Orientation to planes of motion and directional terms for the dog. (From Dyce KM: *Textbook of veterinary anatomy,* ed 4, St Louis, 2010, Saunders.)

Dogs have much more limitation in motion in the dorsal and transverse planes.

## Bones

The bones of the dog skeleton and limbs are illustrated in Figures 5-2, 5-3, and 5-4. Bony landmarks on the bones of the limbs are shown in Figures 5-5 through 5-10.

## Forelimb

The forelimb skeleton consists of the thoracic or pectoral girdle and bones of the forelimb (see Figures 5-5 and 5-6). The size of forelimb bones varies a great deal, because of the greater variation in size for breeds of dogs. The forelimbs bear 60% of the dog's weight.

The canine scapula is positioned close to the sagittal plane. Dogs have an abbreviated clavicle that does not articulate with the rest of the skeleton. It is a small oval plate often 1 cm or less in length and ⅓ cm wide, located at the tendinous intersection of the brachiocephalicus muscle. The adult canine clavicle is mostly cartilage and is usually not visible on radiographs.

The canine humeral head is less rounded compared with the human head, to assist with weight bearing. Distally, there is an olecranon fossa and supratrochlear foramen for the secure positioning of the protruding anconeal process of the ulna for more stability in weight bearing.

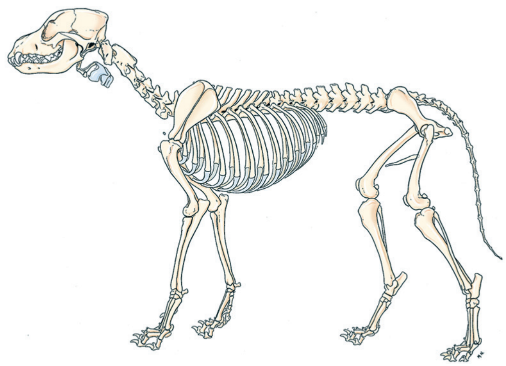

**Figure 5-2** Skeleton of a male dog, left lateral view. (From Evans HE: *Miller's anatomy of the dog,* ed 4, Philadelphia, 2013, WB Saunders.)

The radius is the medial forearm bone and is the main weight-bearing bone of the antebrachium distally. The proximal surface of the radius articulates with the humeral capitulum, which is not as prominent as in the human. The canine distal radius has distinct facets for articulation with carpal bones, providing stability in weight bearing.

The ulna is the lateral forearm bone and has a very prominent olecranon process, which allows secure attachment for the large triceps brachii muscle, needed as an antigravity muscle for weight bearing in dogs. The ulna is the longest bone of the canine body. It articulates distally with the ulnar carpal and accessory carpal bones by two distal facets and does not have an articular disk. The dog has an anconeal process, which is near the attachment site of the anconeus muscle. The anconeal process is needed for stability in weight bearing.

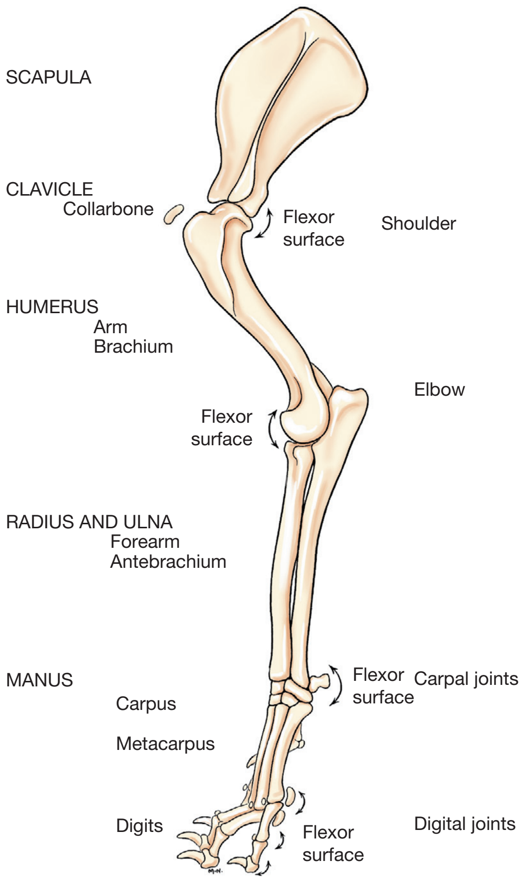


**Figure 5-3** Left forelimb skeleton, noting joints and flexor surfaces. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

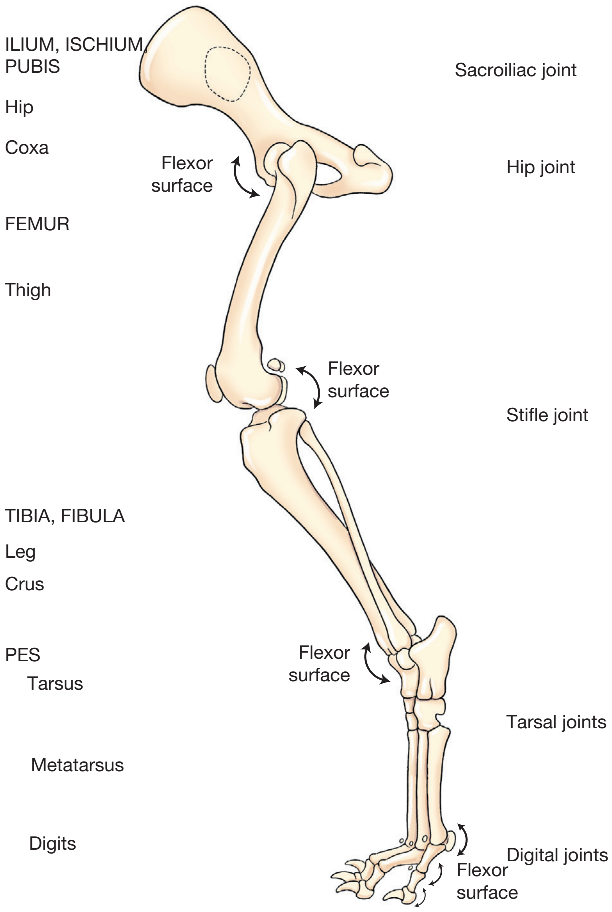


**Figure 5-4** Left hindlimb skeleton, noting joints and flexor surfaces. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

At the carpus or wrist (see Figure 5-7), there are seven carpal bones. The radial carpal bone is analogous to the fused scaphoid and lunate. There are five metacarpal bones. The first metacarpal is short and nonfunctional. Dogs have many sesamoid bones that are embedded in tendons or near them. Sesamoid bones occur when there are significant changes in directions of pull on tendons in addition to the tensile forces produced during muscle contractions. They allow for constant, biomechanically advantageous alignment of angles of insertion of tendons at their attachment sites, which helps relieve stress on the tendinous insertions for animals that walk on their digits. Dogs are digitigrade animals and bear weight on digits II to V, with the main weight bearing occurring on digits III and IV. The sesamoid bones at the dorsal surface of each metacarpophalangeal joint align the extensor tendons for optimal muscle action. Those on the pad surface of the manus align the flexor tendons.

## Hindlimb

The hindlimb skeleton includes the pelvic girdle, consisting of the fused ilium, ischium, and pubis, and the bones of the hindlimb (see Figures 5-8 and 5-9). The size of hindlimb bones varies a great deal, because of the great variation in size for breeds of dogs. The hindlimbs bear 40% of the dog's weight.

The canine pelvis is positioned between the dorsal and transverse planes and closer to the dorsal plane. The canine pelvis is relatively small and narrow. The canine ischiatic or ischial tuberosities are wide and project caudally to form a broad ischiatic table. The canine pelvis shape from a ventral view resembles a rectangle. The symphysis pelvis is relatively long and has two portions, the symphysis ischii and symphysis pubis, compared with the relatively shorter joining of the anterior aspect of the human innominates at the symphysis pubis. The distinction of the shape of the male and female pelvic inlet and outlet in humans is not made in dogs.

The canine femur is the heaviest[4] and largest[5] canine bone. In most dogs, it is slightly shorter than the tibia and the ulna and approximately one-fifth longer than the humerus. The average canine angle of inclination or cervicofemoral angle is 144.7 degrees.[5] Dogs have an average

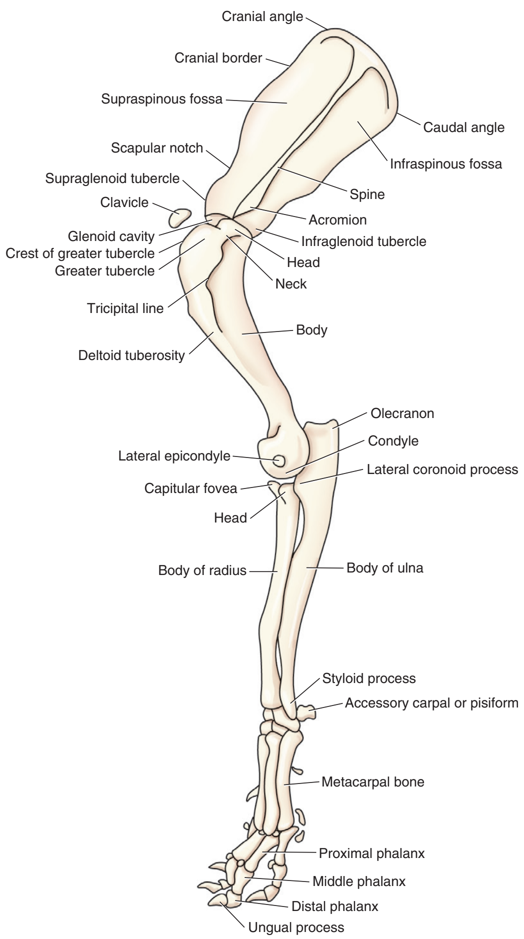


**Figure 5-5** Skeleton of the lateral forelimb of the dog. (Adapted from Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

degree of anteversion or positive femoral torsion of +27 to 31 degrees, when measured from a direct radiograph or with a method using trigonometry and biplanar radiography, respectively.[5] The canine femur has a relatively thick and short femoral neck, a caudomedially located lesser trochanter, a prominent lateral greater trochanter, and a relatively short and wide shaft with a narrow isthmus in the middle. The greater trochanter has a craniolateral prominence called the *cervical tubercle.* Dogs have a third trochanter, which is the attachment site of the superficial gluteal muscle. Canine medial and lateral femoral condyles are equally prominent, but the articular surface of the medial femoral condyle projects more cranially than that of the lateral femoral condyle.

There are three sesamoid bones in the caudal stifle joint region. Two are located in the heads of the gastrocnemius muscle caudal to the stifle joint and are called *fabellae.* The sesamoid in the lateral head is the largest, is palpable, and articulates with the lateral femoral condyle, whereas the one in the medial head is smaller and may not have a distinct facet on the medial femoral condyle. The third is the smallest, is located in the proximal attachment of the popliteus muscle, and articulates with the lateral tibial condyle.

The canine patella, or kneecap, is the largest sesamoid bone in the body. It is an ossification in the quadriceps femoris muscle. The patella alters the pull, increases the moment arm, and protects the quadriceps tendon, as well as provides a greater contact surface for the tendon on the trochlea of the femur than would exist without the patella. The canine patellar articular surface is mildly convex.

The canine tibia is the major bone in the crus. The triangular proximal tibia is wider than the distal cylindrical tibia. Medial and lateral tibial condyles, an intercondylar eminence, and a tibial tuberosity are on the proximal tibia. The tibial plateau slopes distally from cranial to caudal. The extensor groove, on the cranial tibia and lateral to the tibial tuberosity, provides a pathway for the long digital extensor muscle. There is a popliteal notch on the caudal tibia in the midline, where the popliteal vessels course. The tibia articulates with the fibula proximally, along the interosseous crest, and distally. The tibial cochlea articulate with the trochlea of the talus to form the talocrural joint.

The canine fibula is a long, slender bone that articulates with the tibia and also serves as a site for muscle attachment. There is a distinctive groove in the lateral malleolus, the sulcus malleolaris lateralis, through which course the tendons of the lateral digital extensor and peroneus brevis muscles.

The tarsus, or hock, consists of the talus, calcaneus, a central tarsal bone, and tarsal bones I to IV (see Figure 5-10). The talus articulates with the distal tibia and has prominent ridges. At the talocrural joint, two convex ridges of the trochlea of the talus articulate with two reciprocal concave grooves of the cochlea of the tibia. The orientation of the grooves and ridges deviates laterally approximately 25 degrees from the sagittal plane. This deviation allows the hindpaws to pass lateral to the forepaws when dogs gallop.[4] The calcaneus is large and serves as the insertion of the common calcaneal tendon. The central tarsal bone lies between the talus and the numbered tarsal bones I to III. Tarsal IV is large and articulates with the calcaneus and metatarsal bones, spanning this entire region.

The canine hindpaw has five metatarsal bones; however, the first metatarsal can be short or absent. Dogs have many sesamoid bones that are embedded in tendons where there are significant compressive and tensile forces produced during muscle contractions. The sesamoid bones at the dorsal surface of each metatarsophalangeal joint align the extensor tendons for optimal joint action. The sesamoid bones on the plantar surface of the hindpaw align flexor tendons.

## Spine

The spine consists of five areas of the vertebral column: the cervical vertebrae and its articulation with the head,

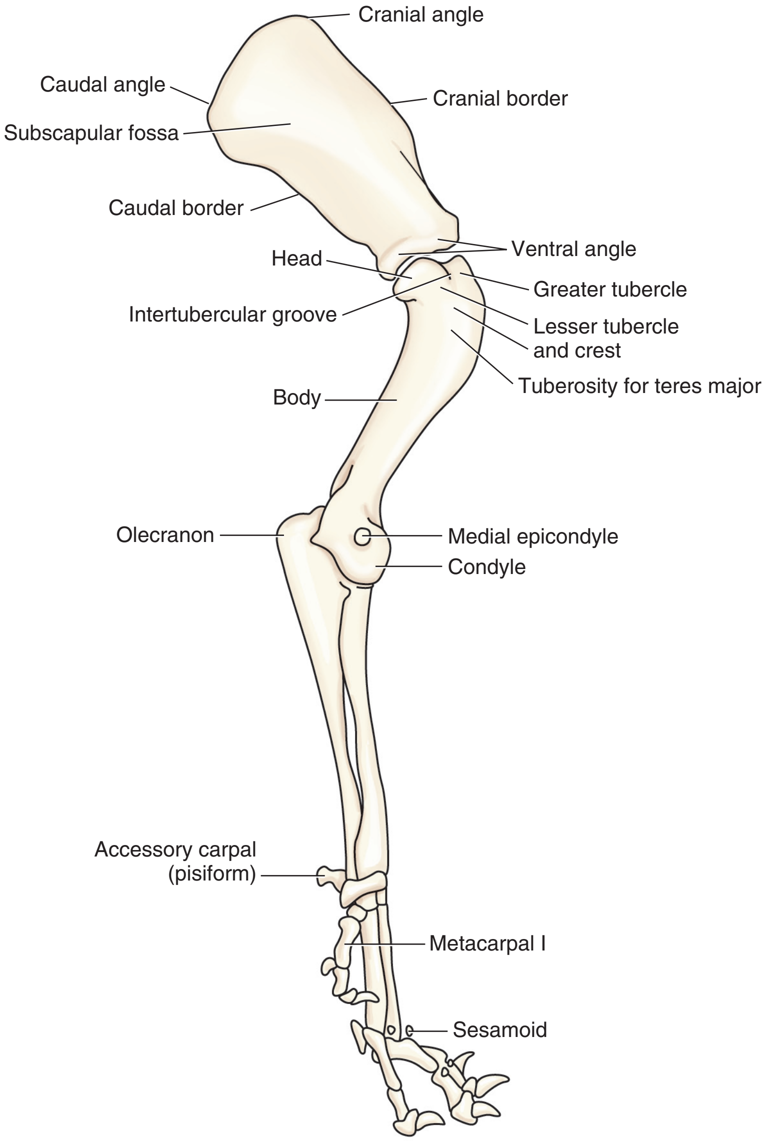


**Figure 5-6** Skeleton of the medial forelimb of the dog. (Adapted from Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

thoracic vertebrae, lumbar vertebrae, sacral vertebrae, and the coccygeal vertebrae (Figures 5-11 through 5-14). The number of vertebrae is listed in Box 5-1.

All vertebrae, except the sacral vertebrae, remain separate and form individual joints. Four sites with limited motion exist within the canine spine.[6] These sites occur at areas where the cranial and caudal articular surfaces are inclined in a nonparallel manner and in different directions. The nonparallel alignment of the articular surfaces markedly restricts joint accessory motions, such as glides. The restricted joint motions and areas resulting from these joint alignments include atlantoaxial motion other than rotation, the cervical (C) 7-thoracic (T) 1 junction, the caudal thoracic region, and the sacrum.

Individual vertebral bone size and shape vary among breeds. For any one breed, canine cervical through lumbar vertebrae are fairly consistent in size. The consistent size in dogs reflects the relatively equivalent cranial-to-caudal compressive loading. Because dogs are quadruped, there is weight bearing on all four limbs. There is cervical spine compression as a result of the positioning of the dog's head as a cantilever, which requires cervical extensor muscle activity to maintain head posture. The massive cervical extensor muscle activity requires relatively large and strong cervical vertebrae to support the muscle mass. Canine intervertebral disks likewise change little in size from the cervical through the lumbar vertebrae. The C5-C6 area is a site of relative hypermobility in large dogs. The spinal cord ends at lumbar (L) L6-L7.

The canine atlas, or C1 vertebra (see Figure 5-12), has a transverse foramen in each transverse process, a craniodorsal arch, and right and left lateral vertebral foramina

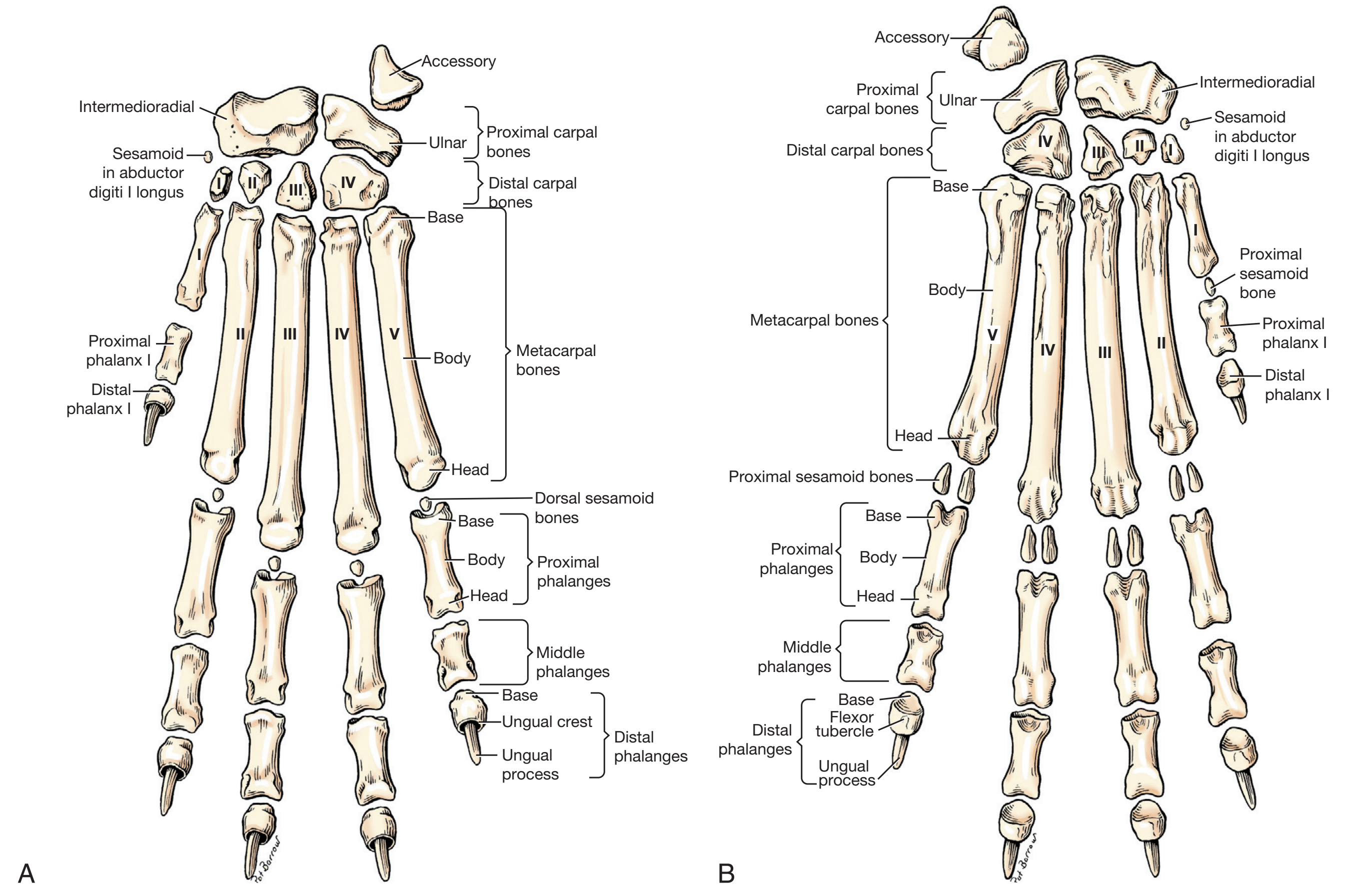


**Figure 5-7** Skeleton of the left dorsal (*A*) and left palmar (*B*) forepaw of the dog. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

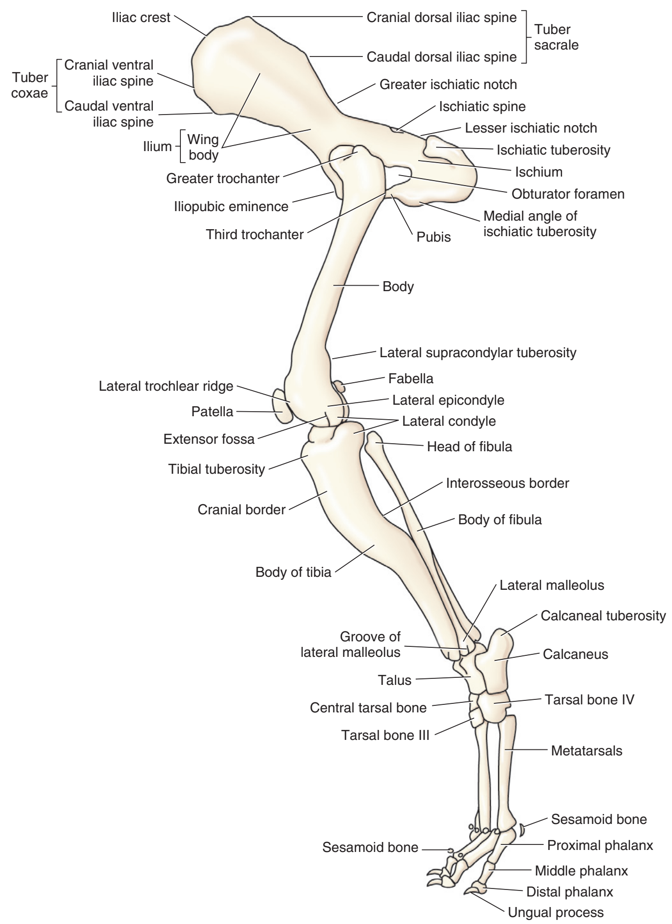


**Figure 5-8** Skeleton of the lateral hindlimb of the dog. (Adapted from Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

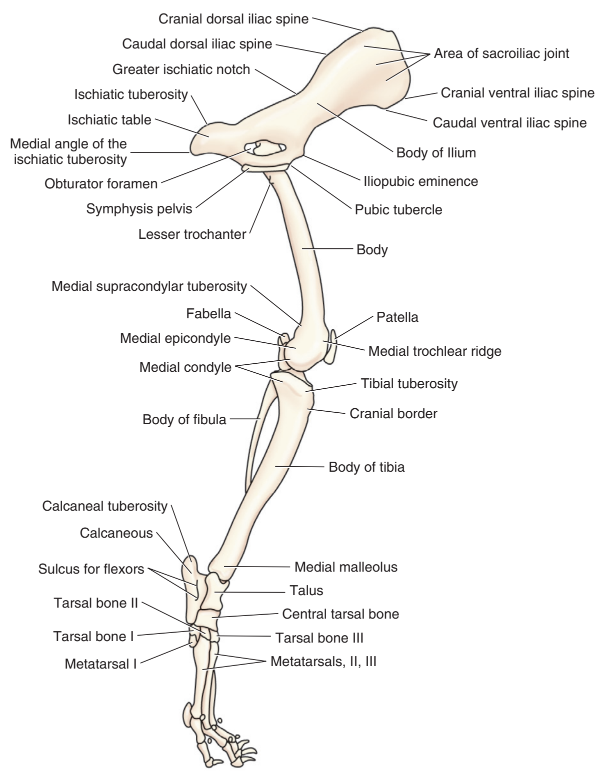


**Figure 5-9** Skeleton of the medial hindlimb of the dog. (Adapted from Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

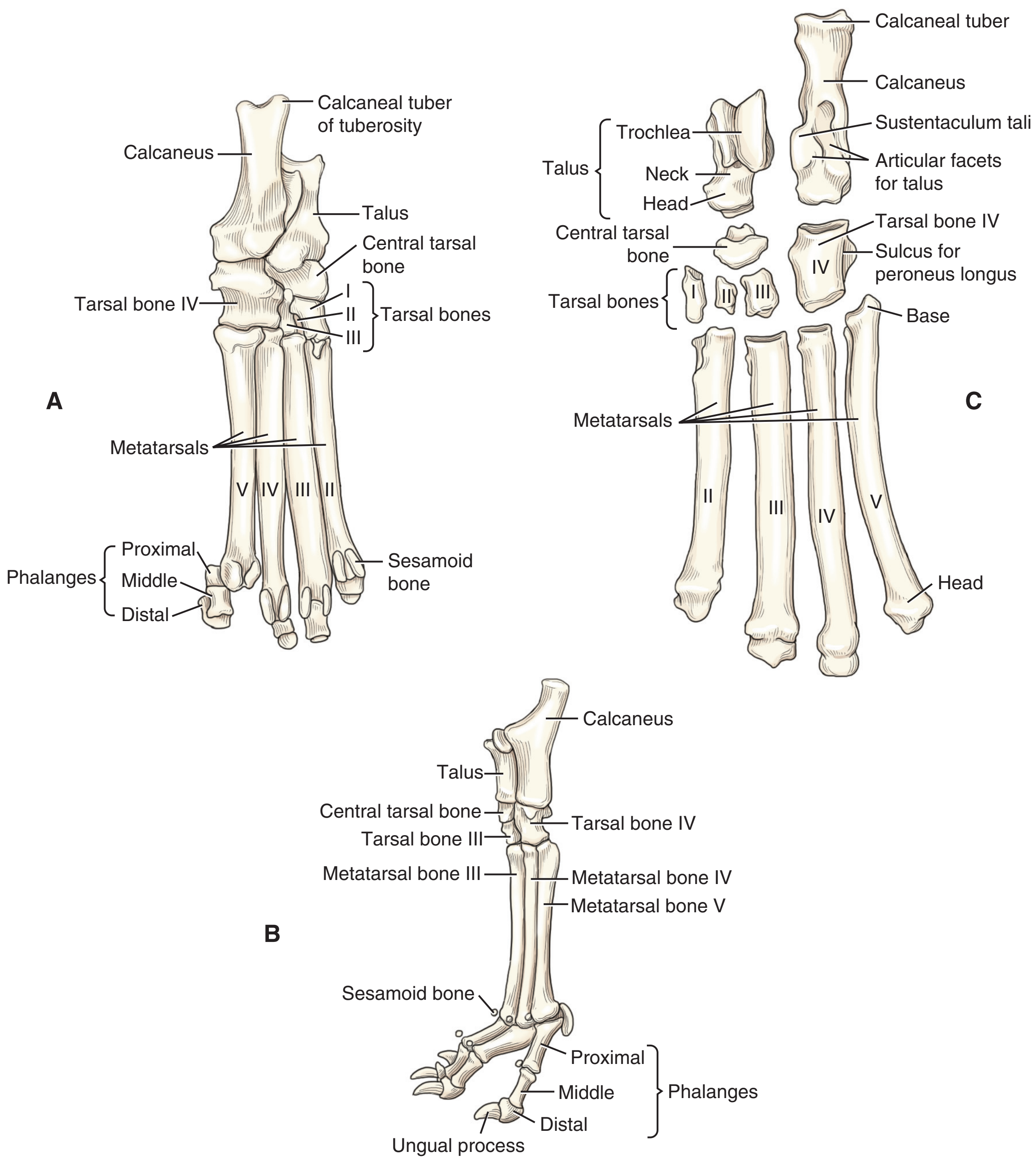


**Figure 5-10** Skeleton of the left plantar (*A*), left lateral (*B*), and left dorsal (*C*) hindpaw of the dog.

for the passage of cervical spinal nerve 1. The atlas has correspondingly shaped condyles for articulation with the occiput. The canine lateral wings or transverse processes are prominent and easily palpable from the skin surface. The canine axis or C2 has a large spinous process with an expanded arch, a wide body, and large transverse processes (see Figure 5-12). The spinous process is nonbifid. The canine axis is very large relative to the size of other canine cervical vertebrae. The axis has a dens, which projects cranially to allow pivotal motion between the atlas and axis. The condyles are oriented near the transverse plane to allow cervical spine rotation. The C3-C6 vertebrae have nonbifid spinous processes, large and flat spinous processes, caudal and cranial articular surface facets that are narrower than the transverse processes, large transverse processes, and transverse foramina for the passage of vertebral arteries. Caudal and cranial articular surfaces are oriented between the dorsal and transverse planes to facilitate cranial and caudal glides needed for cervical spine flexion and extension. The C7 vertebra has a similar shape, a large prominent nonbifid spinous process, and caudal and cranial articular surfaces, which are oriented nearly craniocaudally.

Thoracic vertebrae (see Figure 5-13) have small bodies relative to the size of the entire vertebrae. Canine spinous

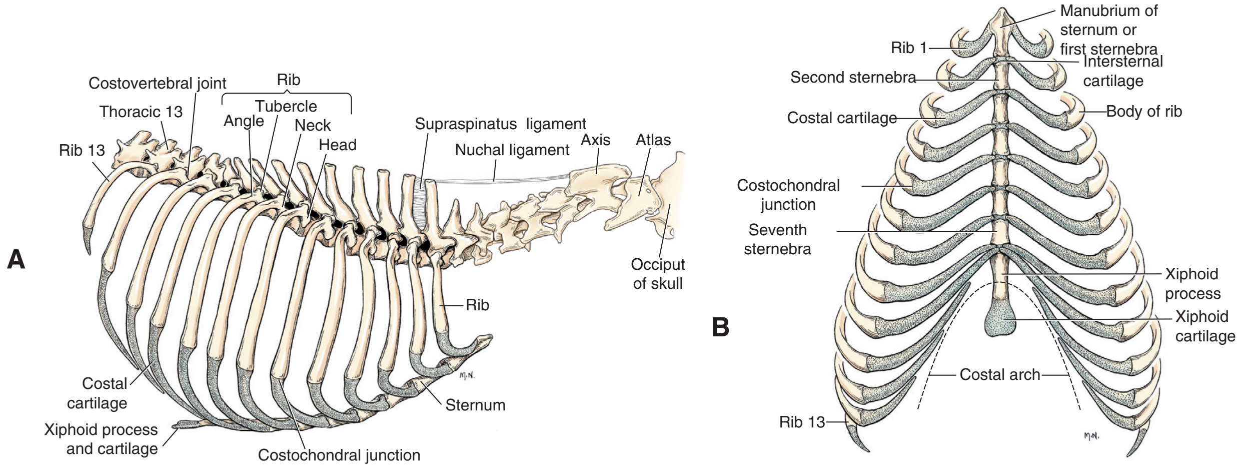


**Figure 5-11** **A,** Identified portions of the axial skeleton cranial to the thirteenth thoracic vertebra. **B,** Ribs and sternum, ventral view.

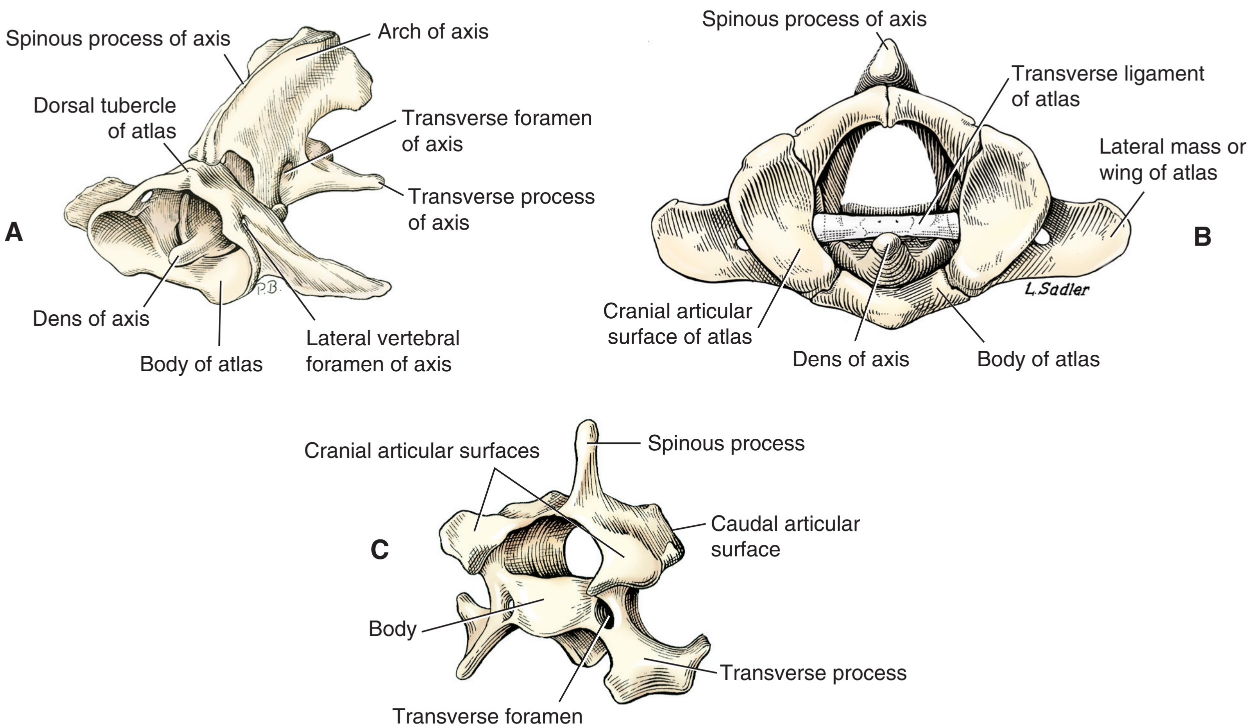


**Figure 5-12** Detailed skeletal anatomy of the atlas and axis from a craniolateral view (*A*), atlas and axis from a cranial view (*B*), and C5 vertebra from a craniolateral view (*C*). (**A** from Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, St Louis, 2010, WB Saunders.)

processes are relatively long. The spinous processes block excessive extension of the thoracic spine. At T10, the size of the body begins to increase and the length of spinous process decreases. The spinous processes are oriented close to the transverse plane. Cranial to T11, the spinous processes project caudally, but caudal to T11, they project cranially. Caudal and cranial articular surfaces are oriented close to the dorsal plane.

Lumbar vertebrae (see Figure 5-13) have bodies that are larger than thoracic vertebral bodies. Canine lumbar transverse processes are long and thin, and they project lateroventrocranially. In the cranial lumbar spine, cranial and caudal articular surfaces are oriented between the transverse and sagittal planes, which facilitate lumbar spine flexion and extension. The L7-S1 joint appears to orient between the sagittal and frontal planes to allow more rotation at this intervertebral level. The canine sacrum is relatively narrow and is linked to the pelvis with sacroiliac joints (see Figure 5-14).

Caudal (Cd) vertebrae (see Figure 5-14) have distinct bodies and transverse processes. The cranial articular surfaces are similar to those in more cranial vertebrae in

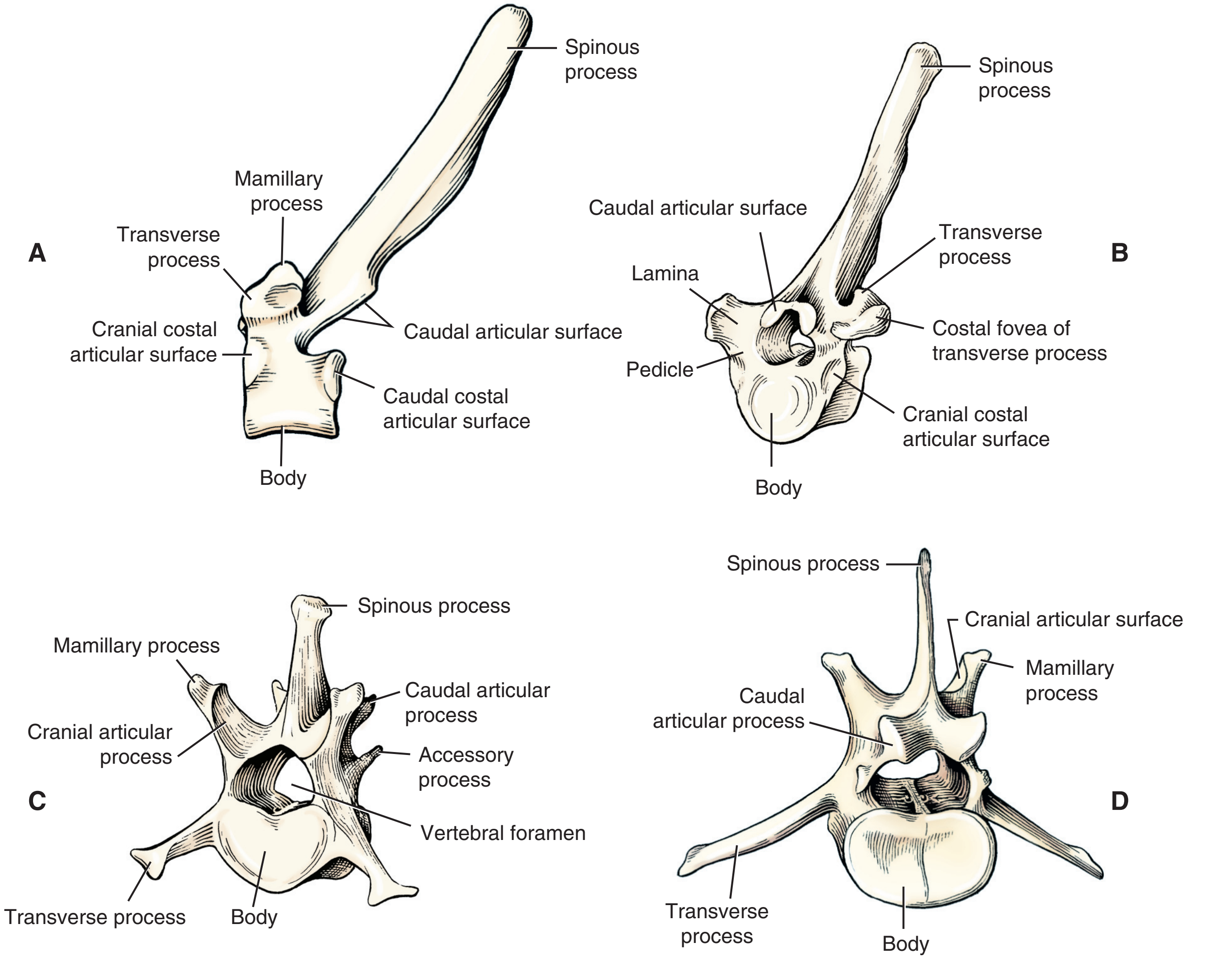


**Figure 5-13** Detailed skeletal anatomy of T6 vertebra from a lateral view (*A*) and craniolateral view (*B*), L1 vertebra from a craniolateral view (*C*), and L5 vertebra from a caudolateral view (*D*).

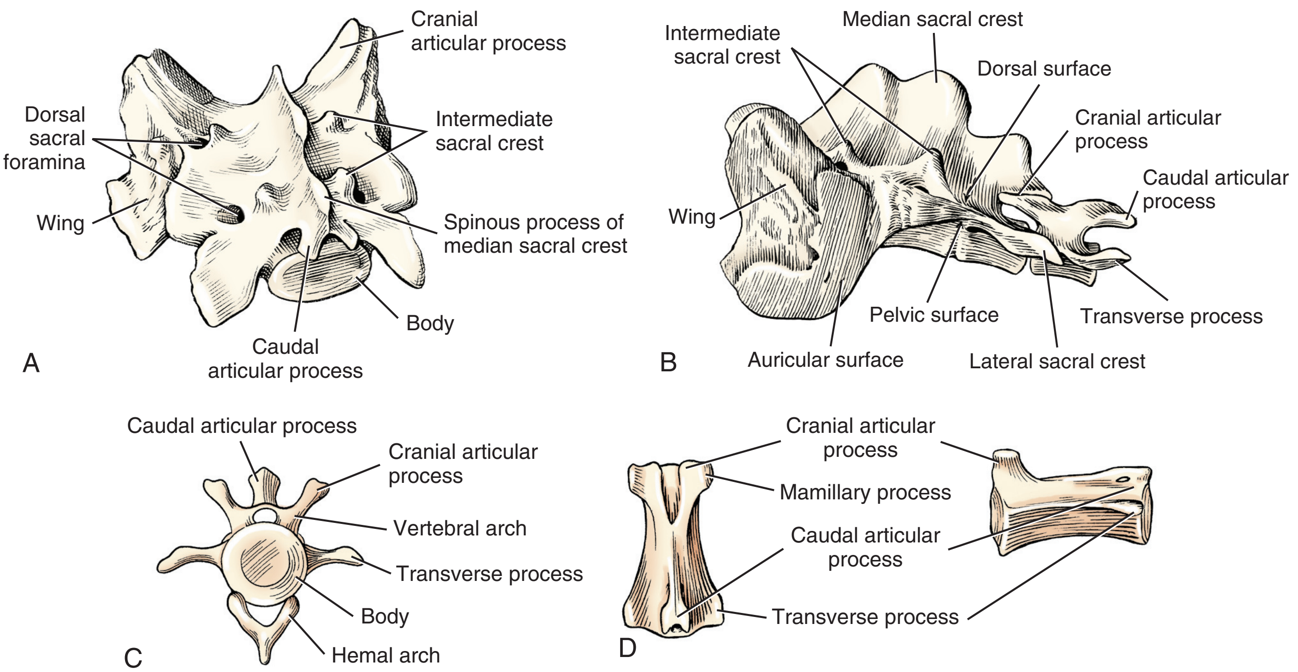


**Figure 5-14** Detailed skeletal anatomy of the sacrum from a caudolateral view (*A*), sacrum and caudal 1 or Cd1 vertebra from a lateral view (*B*), Cd4 vertebra from a cranial view (*C*), and Cd6 vertebra from a dorsal view (*D*). (From Evans HE: *Miller's anatomy of the dog,* ed 4, Philadelphia, 2013, WB Saunders.)

## Box 5-1 Body Segments

### Forelimb

- Forelimb: Arm, forearm, and forepaw
- Thoracic or pectoral girdle
  Scapula, clavicle
- Arm or brachium: Shoulder to elbow
- Forearm or antebrachium: Elbow to carpal joint
- Forepaw or manus
  Carpus or carpals
  Metacarpus or metacarpals
  Phalanges or digits
- Sesamoid bones or cartilages
  Dorsal on MCP joints in common digital extensor tendons of digits II to V; one per digit; small
  Pad surface on MCP joints in interosseous tendons of digits II to V; two per digit; smaller
  Dorsal and palmar on DIP joints of digits I to V; cartilage; small
- One sesamoid bone in the tendon of the abductor pollicis longus
- Digits or phalanges I to V, numbered medial to lateral
  No common names for digits
  Anatomic name: pollex for digit I
- Dewclaw or pollex or digit I with 2 phalanges
- Pads on the paws or digital pads: Weight-bearing pads
  Carpal pad: Small pad palmar to the carpus
  Metacarpal pad: Largest pad palmar to the MCP joints; triangular in shape
  Digital pads: Palmar to the DIP joints; ovoid and flat
- Ungual process: Extension of the phalanx into the claw
- Nails or claws

### Hindlimb

- Hindlimb pelvic limb, or rear limb
  Thigh, leg, hindpaw
- Hip bone or os coxae
  Ilium, ischium, pubis
- Pelvic girdle
  Right and left hip bones and sacrum
- Thigh: Hip to stifle or knee
- Leg or crus: Stifle to talocrural joint
- Hindpaw or hind foot or pes
  Tarsus or tarsals (hock area)
  Metatarsus or metatarsals
  Phalanges or digits or toes
- Sesamoid bones or cartilages
  Dorsal on MTP joints in long digital extensor tendons of digits II to V; one per digit; small
  Plantar surface on MTP joints in interosseous tendons of digits II to V; two per digit; large
  Digit I: One per digit, smaller
  Dorsal and plantar on DIP joints—cartilaginous; one per digit I to V; small
- Digits or phalanges or toes
  I to V
- Dewclaw or digit I or hallux—may be absent, fully developed and articulating with a metatarsal, or may be a vestigial, that is, a trace or rudimentary structure, with a terminal phalanx and no proximal phalanx or metatarsal bone
- Digital pads or pads on the hindpaws—weight-bearing pads
  Tarsal pad: Small pad plantar to the talocrural joint
  Metatarsal pad: Largest pad plantar to the MTP joints; triangular in shape
  Digital pads: Plantar to the DIP joints; ovoid and flat
- Ungual process: Extension of the distal phalanx into the nail
- Nails or claws

### Spine

- Trunk
  Borders: Inguinal ligament to C7-T1 disk
- Neck or cervical spine
- Head
- Tail
- Spinal regions
  Cervical: C1 through C7
  Thoracic: T1 through T13
  Lumbar: L1 through L7
  Sacral: S1 through S3
  Caudal or coccygeal: Cd1-Cd20; some dogs have more or fewer
- Ribs: 13
- Bones in the dog skeleton (excludes auditory ossicles)
  Vertebral column: 50
  Skull: 49
  Hyoid bone: 1
  Ribs: 26
  Sternum: 8 fused bones—manubrium or first sternebra, 6 additional sternebrae, and the xiphoid process
  Forelimbs: 90
  Hindlimbs: 96
  Total: 320
  Other: os penis in males—1
- Hip bone or os coxae
  Ilium, ischium, pubis
- Sacrum
- Pelvic girdle: Right and left hip bones and sacrum
- Pelvic complex: Hip bones, lumbar spine, sacral spine, caudal spine, sacroiliac joints, and hip joints

*C*, Cervical; *Cd*, caudal; *DIP*, distal interphalangeal; *L*, lumbar; *MCP*, metacarpophalangeal; *MTP*, metatarsophalangeal; *S*, sacral; *T*, thoracic.

shape and location; however, the caudal articular processes are bifid and are more centrally located, whereas articular processes in more cranial vertebrae are located more laterally. Hemal arches are separate bones that articulate with the ventral surfaces of the caudal ends of the bodies of Cd4-Cd6. The hemal arches provide protection for the median coccygeal artery, which is enclosed by the arches. In vertebrae caudal to Cd6 and in relatively the same position as the hemal arches are the paired hemal processes, which extend from Cd7-Cd17 or Cd18.

The ribs have vertebral attachments (see Figure 5-11). There are nine pairs of vertebrosternal, or true, ribs and four pairs of vertebrocostal, or false, ribs. The sternum is relatively long and has a manubrium and xiphoid process, with a prominent xiphoid cartilage. The ribs limit overall thoracic spine motion and protect internal organs.

## Joint Motion

The body segments of the forelimb and hindlimb are illustrated in Figures 5-3 and 5-4, respectively, with the major joints and their flexor and extensor surfaces. Body segments are listed and defined in Box 5-1. Types of joints are listed in Box 5-2.

### Joint Motion and Shape of Articular Surfaces

The shape of articular surfaces of bones helps define the motions available for a joint. Articular surfaces of two bones forming a joint are usually concave on one bone and convex on the other bone. Some articular surfaces are flat. Occasionally adjacent bones are convex on both joint surfaces. Intraarticular structures, such as the medial and lateral menisci in the stifle joint, may modify adjacent surfaces. Understanding the concave-convex relationships as a guiding principle in determining joint motion allows prediction of possible joint motions based on articular surface shape. Ligamentous and other soft tissue around the joint guide and restrict the motion that would be possible based on articular surface shape alone.

Joint motions are named, most commonly, by movement of the distal bone relative to the proximal bone. For example, cranial movement of the tibia on a stable femur is named *stifle joint extension.* The major direction of motion, such as flexion of the stifle, is physiologic or osteokinematic motion.

Accessory, or arthrokinematic, motion is smaller in magnitude and less observable. Examples of accessory motions are glide or slide, rotary motion, distraction or traction, and compression or approximation. A normal amount of glide occurs in normal functioning joints. Glides are shear type or sliding motions of opposing articular surfaces. Rolls involve one bone rolling on another. Gliding motion in combination with rolling is needed for normal physiologic joint motion. Spins are joint surface motions that result in continual contact of articular cartilage areas on opposite sides of a joint. Distraction or traction accessory motions are tensile or pulling-apart movements between bones. Compressive or approximation accessory motions are compressive or pushing-together movements between bones.

Normal joint motion involves both physiologic motion and accessory motion. Physiologic motion in joints with opposing concave and convex articular surfaces involves both roll and glide. Roll occurs in the same direction as the movement of the moving segment of the bone, but glide directions differ based on whether the moving articular surface is concave or convex. A glide is described by identifying the joint motion, the direction of the glide, and which bone is moving. For example, stifle flexion involving the tibia and femur is termed *caudal glide of the tibia on the femur.*

### Joint Motion in the Limbs and Spine

Joint motions are named by one body segment approaching or moving away from another body segment or movement of some referenced body landmark. Joint motions are named in the following sections and described (see Figures 5-3 and 5-4) as they refer to the limbs, starting from normal stance. Limb motion is usually described by motion of the joint rather than a body segment. For example, elbow flexion is recommended rather than forearm flexion. Occasionally, body segment motion is used to describe limb motion when motion does not involve axial motion with a joint as a pivot point. For example, rotation of the forelimb might be observable when pronation at the radioulnar joint would be difficult to observe clinically.

#### *Flexion*

During flexion, a limb is retracted or folded, a digit is bent, and the back or neck is arched dorsally (i.e., the convex portion of the arch is directed dorsally). In the limbs, flexion motion occurs as the bones on either side of a joint move closer together and the joint angle becomes more acute. Flexion may also be referenced to limb motions involving closing angles during the swing phase of gait. Flexion motions of the limb joints are noted in Figures 5-3 and 5-4. In the spine, flexion occurs as the back or neck arches dorsally (i.e., the convex portion of the arch is directed dorsally).

A notable difference between dogs and humans is the meaning of *shoulder flexion.* In dogs, caudal retraction of the humerus in relation to the scapula is shoulder flexion, whereas cranial motion of the humerus in relation to the scapula is shoulder extension. The direction of shoulder flexion motion is opposite to this in humans. The terminology used in dogs is consistent with naming flexion as described previously.

In normal stance, as shown in Figure 5-2, a dog's spine is flexed at the atlantooccipital and atlantoaxial joints, straight (neither flexed nor extended) in the remainder of the cervical spine, extended at the cervicothoracic junction, slightly lordotic in the thoracic spine, and flexed or normally kyphotic in the lumbar spine. There is either a slightly flexed or extended sacrum on the lumbar spine, depending on the tail posture. The flexed canine lumbar spine is beneficial to running speed. During running, the lumbar spine moves through varying degrees of flexion as running speed changes.

#### *Extension*

During extension, the limb reaches out, the digit is extended, and the back or neck is less arched dorsally or

**Box 5-2 Types of Joints**

**Forelimb**

- Artificial joint: Not described as a joint
- Ball and socket: Shoulder
- Hinge: Elbow, metacarpophalangeal I
- Pivot: Proximal, and distal radioulnar
- Syndesmosis: Middle radioulnar
- Hinge with lateral motion: Carpal
- Ellipsoid: Antebrachiocarpal, radiocarpal
- Plane: Middle carpal or midcarpal, intercarpal, intermetacarpal
- Saddle plane: First carpal with MC I
- Plane: Second carpal with MC II, third carpal with MC III, fourth carpal with MC IV and V intermetacarpal
- Hinge: Metacarpophalangeal I
- Condylar or condyloid: MC II to V with the same numbered proximal phalanx
- Saddle/condylar
  Proximal interphalangeal II to V
  Distal interphalangeal II to V
  (Interphalangeal of thumb)

**Hindlimb**

- Synovial and fibrous: Sacroiliac
- Symphysis: Symphysis pelvis
- Ball and socket: Hip or coxofemoral
- Complex condylar: Stifle (the term *knee* is used commonly with an animal's owner)
- Plane
  Patellofemoral
  Tarsal joints or hock joints (this joint is referred to as the *hock joint* in common usage)
    Talocalcaneal
    Talocalcaneocentral and calcaneoquartal joints combined
    Proximal intertarsal or talocentral
    Calcaneoquartal
    Calcaneocentral
    Centroquartal
    Centrodistal
    Distal intertarsal: Central bone with tarsal III
    Tarsal I with II, II with III
    Tarsal III with IV
  Tarsometatarsal
    Tarsal I with MT I
    Tarsal II with MT II
    Tarsal III with MT III
    Tarsal IV with MTs IV and V
  Intermetatarsal
- Tibiofibular
  Synovial: Proximal and distal tibiofibular
  Syndesmosis: Middle tibiofibular
- Hinge: Talocrural, tarsocrural, tibiotarsal (the tarsocrural has been referred to as the talocrural and the talocalcaneal joints combined) or ankle joint (the term *ankle* is commonly used with an animal's owner)
  Metatarsophalangeal I—MT I with digit I
- Condylar: MT II to V with the same numbered digit
- Saddle
  Proximal interphalangeal II to V
  Distal interphalangeal II to V
  Interphalangeal of hallux

**Spine**

- Part synovial and part fibrous: Sacroiliac
- Condyloid: Atlantooccipital
- Pivot: Atlantoaxial—dens of C2 and atlas
- Plane
  Atlantoaxial—articular surfaces
  Between cranial and caudal articular surfaces
  Costovertebral
  Sternocostal: Sternum and true ribs
  Synchondrosis: Costochondral—ribs with cartilage

*MC*, Metacarpal; *mT*, Metatarsal.

arched ventrally. Extension is motion in the sagittal plane in the direction opposite to that of flexion motion. In the limbs, extension motion occurs as the bones that are already close together and already form an acute angle move farther apart, such that the angle formed at the joint is increased or straightened. Extension beyond normal is sometimes termed *hyperextension*. The carpus normally has greater than 180 degrees of extension. In the spine, extension occurs as the back or neck is arched ventrally (i.e., the convex portion of the arch is directed ventrally).

### *Abduction*

Limb joint abduction is movement in the transverse plane such that the distal aspect moves away from the midline of the body. Abduction of the digits is not routinely measured in dogs. Separation of the toes may be noted as splayed toes or abducted toes. The reference line for abduction of the digits passes between digits III and IV. Digit I, or the dewclaw, in dogs is nonfunctional and is not considered in a discussion of skeletal motion.

### *Adduction*

Limb joint adduction is movement in the transverse plane such that the distal aspect moves toward the midline of the body. Adduction of the digits is not routinely measured in dogs. The reference line for adduction is the same as for abduction.

### *Side Bend*

In the trunk and head, side-to-side movement occurs in the dorsal plane such that lateral aspects of adjacent areas of the spine approximate or move away from each other. For example, side bend to the right involves the right side of adjacent spinal areas approximating each other.

### *Circumduction*

Circumduction is movement at a joint, during which a bone or body segment outlines the surface of a cone or circle. This motion does not require rotation at the joint, but rotation at the joint may occur. Forelimb circumduction occurs primarily at the shoulder. Hindlimb circumduction occurs at the hips. Although circumduction at individual limb joints is possible in dogs, veterinarians commonly observe and document circumduction of the entire limb, without distinction of the joints that contribute to the overall motion.

### *Rotation*

Internal rotation of a limb occurs with reference to the cranial aspect of the limb rotating toward the midline of the body or median plane and continuing in that direction. External rotation of a limb occurs with reference to the cranial aspect of the limb rotating away from the midline of the body and continuing in that direction. Limb rotational motions occur in the dorsal plane. With reference to the ventral aspect of a vertebral body, rotation to the right or left is called *rotation to the right or left,* respectively. Spinal rotations occur in the transverse plane.

### *Supination*

In the limbs, supination is external rotation of the limb, such that the pad surface of the paw faces medially. Supination motion therefore could involve forelimb carpal and forepaw supination, shoulder external rotation, or hindlimb talocrural and hindpaw supination, as well other motions such as hip joint external rotation. Scapular motion and glenohumeral external rotation occurring as part of supination occur in the dorsal plane. However, dorsal plane scapular motion is blocked by contact of the scapula with the trunk.

### *Pronation*

In the limbs, pronation is internal rotation of the limb, such that the pad surface of the paw faces laterally. Pronation motion therefore could involve forelimb carpal and forepaw pronation, as well as other motions, opposite to those of supination.

### *Other Shoulder Girdle Motions*

Other shoulder girdle motions that are defined for humans exist in dogs, but have not been defined and measured in dogs. They include the human motions of scapular protraction, retraction, upward rotation, downward rotation, elevation, depression, anterior tilt and return from anterior tilt of the scapula, and glenohumeral horizontal abduction and horizontal adduction. Human scapulohumeral rhythm has not been defined in dogs, but may exist.

### *Pelvic and Sacroiliac Motions*

Pelvic and sacroiliac motions and postures are less well defined in dogs than in humans. Human pelvic and sacroiliac motions and postures are described in depth, although terminology is not consistent among authors and clinicians. When pelvic and sacroiliac motions and postures are analyzed and described in dogs, the following terminology could be useful for correct identification and for accurate communication. Pelvofemoral rhythm, the human lower-limb counterpart to human scapulohumeral rhythm, has not been defined in dogs, but may exist.

## Pelvic Motion

### *Ventral and Dorsal Pelvic Tilt*

Orientation of the pelvis with respect to the sagittal plane describes tilt of the pelvis. Ventral tilt involves approximation of the ventral pelvis and cranial femur, and dorsal tilt involves the ventral pelvis moving away from the cranial femur. These motions occur with the hip joints as the axes of rotation. Relative to a longitudinal axis through the entire spine, the canine pelvis appears tilted dorsally on the canine longitudinal axis.

### *Lateral Tilt of the Pelvis*

Lateral tilt is dorsal plane movement such that the lateral aspect of the pelvis approximates the lateral aspect of the trunk. Lateral pelvic tilt to the right, for example, would occur as the cranial border of the pelvis tilts to the right, approximating the right lateral side of the pelvis toward the right trunk.

### *Pelvic Rotation*

With reference to the movement of the ventral aspect of a pelvis, rotation toward the right or left is called *rotation to the right or left,* respectively. This motion occurs in the transverse plane. Pelvic rotation can be referenced to the floor and one lower limb. If this convention were used, internal rotation with reference to the right hindlimb would occur as the ventral pelvis rotates to the left. Internal rotation motion of the right hindlimb would occur in the same direction, that is, with the cranial aspect of the thigh or femur rotating to the left and the ventral aspect of the leg or tibia and fibula rotating to the left.

### *Sacroiliac Motion*

The suggested but as yet undefined terminology uses application of human terminology, with movements of the canine cranial and caudal aspects of the sacrum correlated to movement of the human superior and inferior aspects of the sacrum, respectively.

### *Sacroiliac Flexion or Nutation*

Sacroiliac flexion involves the cranial aspect of the sacrum tilting ventrally.

### *Sacroiliac Extension or Counternutation*

Sacroiliac extension involves the cranial aspect of the sacrum tilting dorsally.

### *Sacroiliac Side Bend or Lateral Flexion*

Sacroiliac lateral flexion involves the cranial aspect of the sacrum tilting to the side. Right side bend or lateral flexion occurs as the cranial aspect of the sacrum tilts to the right.

### *Sacroiliac Rotation*

Sacroiliac rotation involves rotation around a longitudinal axis through the sacrum. Rotation of the sacrum is identified by the motion of the ventral sacrum to be consistent with terminology for humans. For example, rotation of the ventral sacrum to the left is called *sacral rotation to the left.*

Overall spinal motion is determined and affected by many things. Among them are the orientation of the articular surfaces (facets); disk shape and size; structural blockages, such as ribs and orientation of spinous processes; ligament fiber direction, strength, and tightness; and muscle-tendon unit strength and tightness.

## Normal Joint Motions and Ranges of Motion

Appendix 2 summarizes normal joint motions and ranges of motion for the peripheral joints.[5,7,8] Joint ranges of motion for the spine have not been documented in dogs.

## Distinction of Maintained Joint Postures versus Joint Motion

Motion is described by either referencing the joint or referencing a body segment. Terms to describe motion include *flexion, flexing, abduction, abducting,* and so on. Motion occurs in a consistent direction, independent of the starting point of the motion. For example, the movement of the carpal joint from neutral to flexion is carpal joint flexion. Movement of the carpal joint from full extension to neutral is carpal joint flexion. Postures are maintained positions. Terms to describe postures include *flexed* or *abducted, maintained flexion,* or *abduction.*

# Skeletal Alignment

## Limbs

Skeletal alignment is important, because it is closely linked to mobility, function, and pathologic conditions. A functional example of the importance of skeletal alignment is seen at the talocrural joint. Alignment at this joint influences angulation of the hindlimbs in relation to speed during the gallop. Figures 5-5, 5-6, 5-8, and 5-9 show the normal alignment of the limbs.

Skeletal alignment is the alignment between bones at joint surfaces or the alignment of one portion of a bone in relation to another portion of the same bone. *Skeletal malalignment* is abnormal alignment of either of these two relationships.[9] Skeletal alignment at any one joint affects the alignment at other joints, especially at joints functioning in weight-bearing or closed-chain activities rather than non–weight-bearing or open-chain activities. Skeletal malalignment can lead to arthritis of a malaligned joint or of a compensating joint.

Normal skeletal alignment results in normal correlated motions and postures, which occur naturally and are normally related to the motion or posture imposed at the joint. Correlated motions and postures may not occur at all but can occur at any or several joints along the chain. This concept is well defined in human physical therapy, although clinicians use different terms to describe it. These concepts have been discussed in relation to dogs, mostly in the context of canine gait and related to correlated motion as a result of a limb with pain or dysfunction, rather than from skeletal malalignment alone.[10] An example of correlated posture or motion is the lateral rotation of the proximal forelimb or hindlimb that occurs when the dog supinates at the paw. Another example is the excessive shoulder or hip adduction and medial positioning of the elbow or stifle joints, which occur when the digits angle or point laterally in the forepaw or hindpaw, respectively.

For any skeletal malalignment or associated abnormal correlated motion or posture, compensatory motions or postures can occur. They occur to normalize mobility, improve cosmetic appearance of the limb, or improve foot contact with the ground. The term *compensation* is used to describe abnormal motion and posture in dogs.[10] An example is excessive weight bearing on one limb when dysfunction exists in the other limb. Another example of a compensatory motion is limb circumduction while advancing that limb during gait, as a result of inability to flex the joints of a limb during swing.

## Spine

Complex correlated and compensatory postures and motions occur within the spine and pelvic complex. The pelvic complex for dogs and humans is defined in Box 5-1. Human correlated and compensatory postures and motions related to the spine and pelvis are described in great detail in the physical therapy literature[11-14]; however, there is controversy and even contradiction related to which motions and postures are correlated or compensatory for other motions and postures. This concept has been applied in less detail to spinal and pelvic postures and motions in dogs; however, development and use of terminology are encouraged for analysis of spinal motion and posture.

# Fascia (Fasciae)

Fascia (fasciae), formed by flat, enveloping connective tissue, weaves throughout the body and provides structural connection. With damage, it can become adhered to adjacent structures. Physical therapists and other clinicians intervene in such circumstances with treatments such as stretching, transverse friction massage, and soft tissue mobilization to reduce this adherence.

## Forelimb

The superficial and deep fascia of the shoulder and brachium are continuous with the superficial and deep fascia of the cervical spine and thorax. The superficial fascia in the region of the sternum is continuous with the medial brachium, and it covers the cephalic vein during its course. Distal to the elbow, the superficial fascia is more adhered to the deep structures. The cutaneous nerves and vessels of the forelimb travel long distances deep to the superficial fascia before entering the skin.

The deep fascia of the lateral shoulder and brachium, called the *fascia omobrachialis lateralis,* covers the muscles' superficial surfaces and adheres to the spine of the scapula. The deep antebrachial fascia forms a single dense sleeve around the muscles in the caudal antebrachium and is thickest medially. Intermuscular septa extend to the radius and ulna to separate the flexor and extensor carpal muscle groups.

The deep fascia is closely joined with the connective tissue around the tendons of the carpal joint. A thick and strong extension, the flexor retinaculum, bridges the flexor tendons to the carpal joint and digits. On the palmar surface of the metacarpophalangeal joints, thick extensions attach to the sesamoid bones and are called *transverse ligaments.*

## Hindlimb

The hindlimb also has superficial and deep layers of fascia. At times the fascial layers are separable, especially with a layer of fat in obese dogs, and at times they are not. In general, the deep fascia is stronger than the superficial.

The superficial fascia on the dorsal trunk is continuous with the superficial gluteal fascia of the hip area, whereas the superficial fascia on the ventrolateral abdomen is continuous as the superficial lateral fascia of the lateral thigh and gluteal region. The superficial fascia encompasses the thigh almost to the patella and is closely connected to the deep fascia of the biceps femoris muscle. The superficial and deep fascia are also closely associated near the proximal sartorius muscle, the femoral canal, and the gracilis muscle. The superficial fascia of the crus, tarsus, and hindpaw is similar to that in the forelimb.

The strong lateral femoral fascia, or fascia lata (iliotibial band in humans), is formed on the lateral thigh from the gluteal fascia and the fascia of the tensor fasciae latae muscle and attaches to the lateral femur by an intermuscular septum between the biceps femoris and vastus lateralis muscles. The more loosely adhered medial femoral fascia joins the lateral femoral fascia to envelop the thigh and attach to the medial femur by a septum caudal to the vastus medialis. Both lateral and medial fascial layers attach to the patella and femoral condyles. The fascia extends around both the medial and lateral aspects of the stifle, but it thins as it merges with the patellar ligament and patella. The deep fascia continues in the crus as the crural fascia. The crural fascia has a superficial layer, which extends to the metatarsus, and a deep layer. The superficial and deep leaves are united near the distal tendons of the biceps femoris and semitendinosus, where they contribute to the common calcaneal tendon. The layers are also united to the distal tendons of the semimembranosus and cranial tibial muscles. The deep crural fascia thickens to form the crural extensor retinaculum, which envelops the long digital extensor and cranial tibial tendons, and the tarsal extensor retinaculum, which envelops the long digital extensor muscle. These retinacula ensheath all tendons and superficial muscles of the hindpaw and secure them to the bones in the hindpaw.

## Trunk and Neck

The superficial and deep fascial layers of the neck and head are continuous. Although the deep fascia is stronger than the superficial fascia, it is deeper and more difficult to palpate; therefore the superficial fascia is emphasized here. The superficial fascia of the neck ensheathes the entire neck, is continuous toward the forelimb as the superficial brachial fascia, and is continuous ventrally with the superficial fascia of the trunk in the sternal region. Caudally, the superficial external fascia of the trunk is continuous with the superficial gluteal fascia. At the dorsal midline in dogs, there is no attachment of the superficial fascia to deep structures, allowing the superficial fascia to be lifted in a big fold when the skin is gently lifted. Large quantities of subfascial fat are deposited in areas where the superficial fascia and skin move freely on deeper structures.

The deep external fascia is attached to the supraspinous ligament on the spinous processes of thoracic and lumbar vertebrae and ends cranially on the transverse processes of C7 and T1. In the trunk, it is continuous with the lateral abdominal muscles. The deep fascia attaches to the sternum and costal cartilages cranially and the ilium caudally. The dorsally located thoracolumbar fascia has two layers and lies deep to large amounts of fat in well-nourished dogs. The main layer serves as an attachment site for the serratus dorsalis cranialis and the internal and external abdominal oblique muscles. The deep fascia is continuous with the deep gluteal fascia and other deep fascia of the thigh.

## Compartments

Fascial layers and the bones they attach to can form well-defined volumetric spaces called *compartments*. Compartments are described in the limbs and contain muscles and their associated neurovascular supply. Compartments become important when interstitial fluid hydrostatic pressure within the compartment exceeds physiologic levels, causing ischemia, disruption of membrane integrity, edema, and loss of viable muscle tissue.

Four distinct compartments have been described in the dog: (1) craniolateral compartment of the crus, (2) caudal compartment of the crus, (3) caudal compartment of the antebrachium, and (4) femoral compartment.[15] Unlike the others, the femoral compartment is not a single space but consists of three fascial envelopes; one contains the quadriceps muscles, another the hamstring group, and the final one surrounds the adductor muscle.

# Muscles

Muscles are characterized as primary movers (i.e., critical in performing joint actions) and secondary movers (less important in performing joint actions). Primary movers are the muscles that produce the necessary moments for normal (physiologic) joint motion and have the best combination of characteristics: large physiologic cross-section, large moment arm, and a good line of action based on the desired joint motions. Primary movers usually cross the joint with the desired motion at the last joint crossed.

Muscle strength depends on the multiplied product of the muscle force applied in the direction of the desired joint motions and the moment arm of the muscle force. The size of the muscle and the direction of its line of action force vector determine its ability to produce specific motion. Muscle force is related to the physiologic cross-sectional areas of muscle fibers. The line of action determines the length of the moment arm, which determines the moment arm's contribution to the muscle moment that can be produced.

Interestingly, some canine muscles have been named according to the number of heads in analogous human muscles, but the actual number of heads making up the muscle may be different. For example, the biceps brachii in dogs has one head and the human biceps brachii has two heads, but both muscles are called biceps brachii.

## Forelimb

Muscle number, size, and location are related closely to forelimb function as a weight-bearing limb (Figures 5-15 to 5-21). In addition, the need to suspend the head and neck increases the need for muscle mass and strong ligaments that extend between the head, neck, and forelimb. The

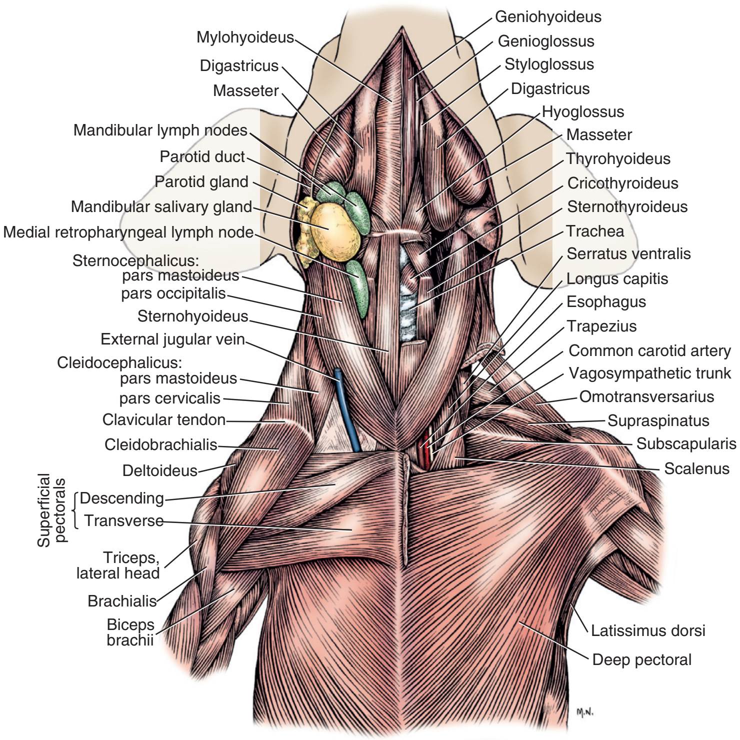


**Figure 5-15** Superficial muscles of the ventral shoulder and neck. (From Evans HE: *Miller's anatomy of the dog*, ed 4, Philadelphia, 2013, WB Saunders.)

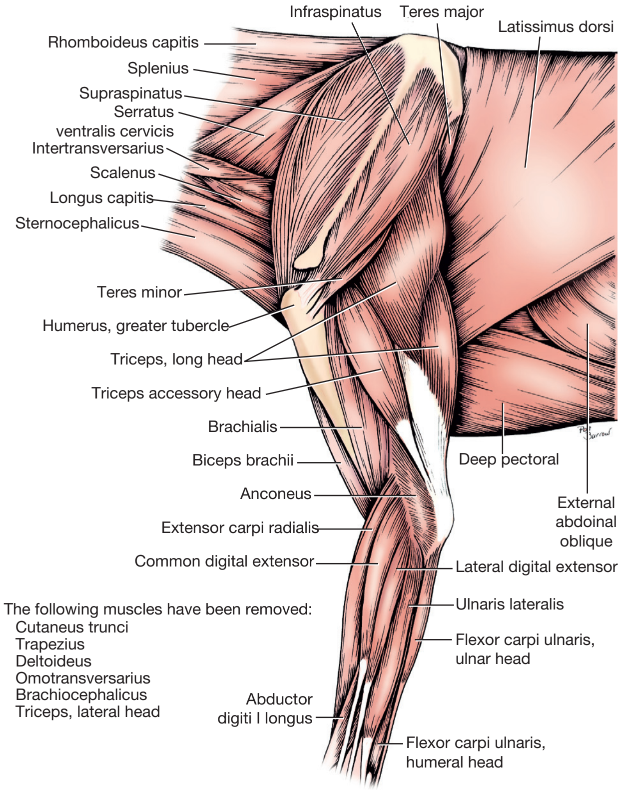


**Figure 5-16** Deep muscles of the left lateral forelimb. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

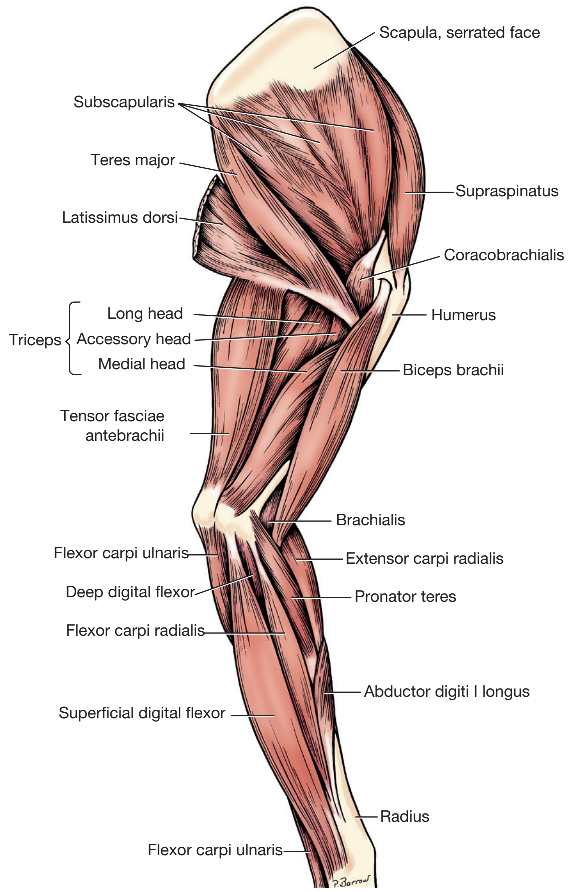


**Figure 5-17** Muscles of the left medial forelimb. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

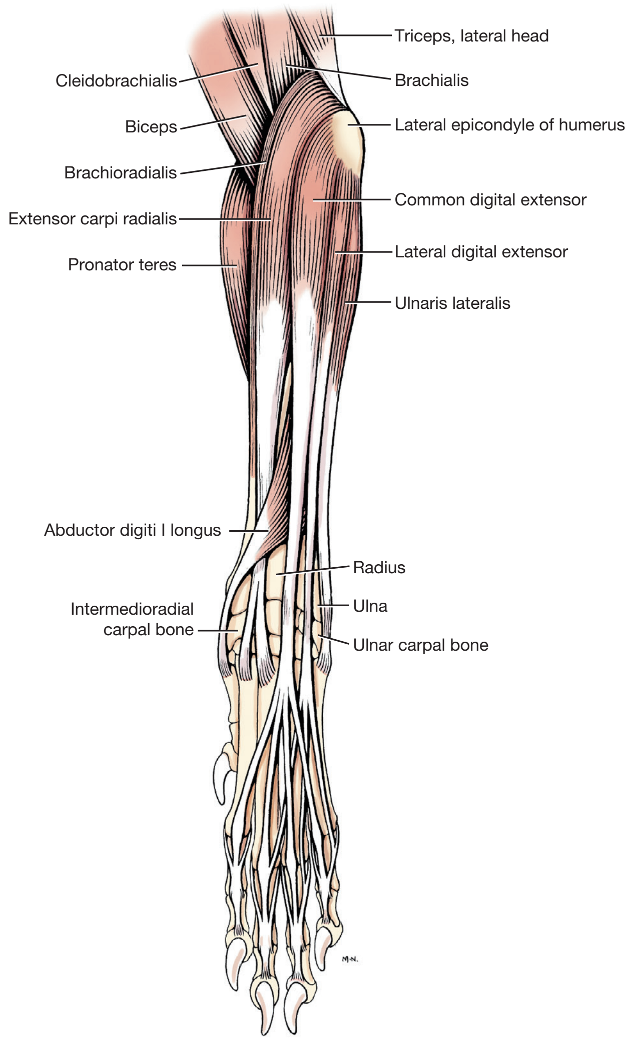


**Figure 5-18** Muscles of the left antebrachium and forepaw, cranial view. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

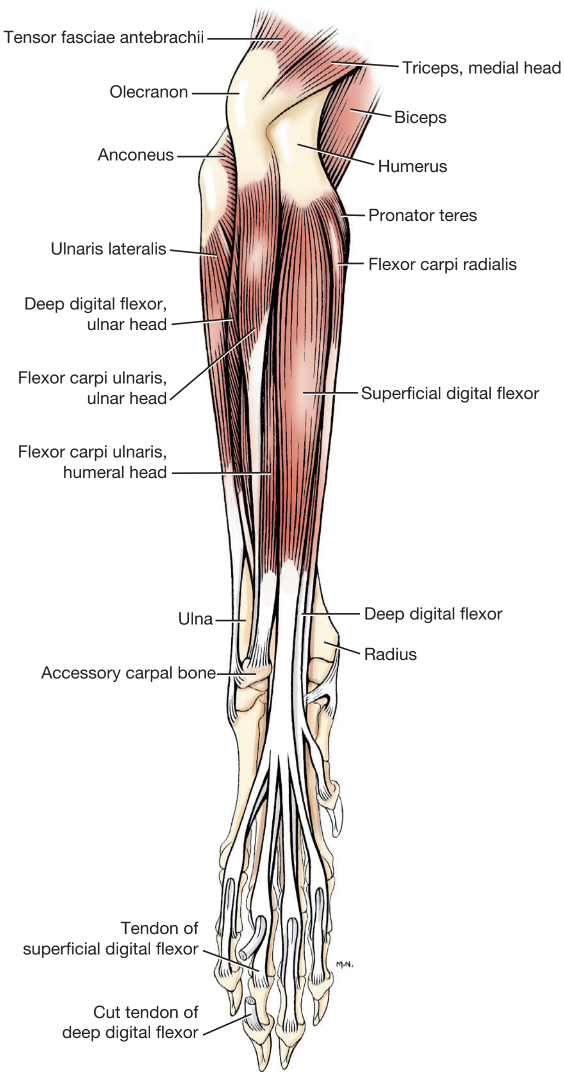


**Figure 5-19** Muscles of the left antebrachium and forepaw, caudal view. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

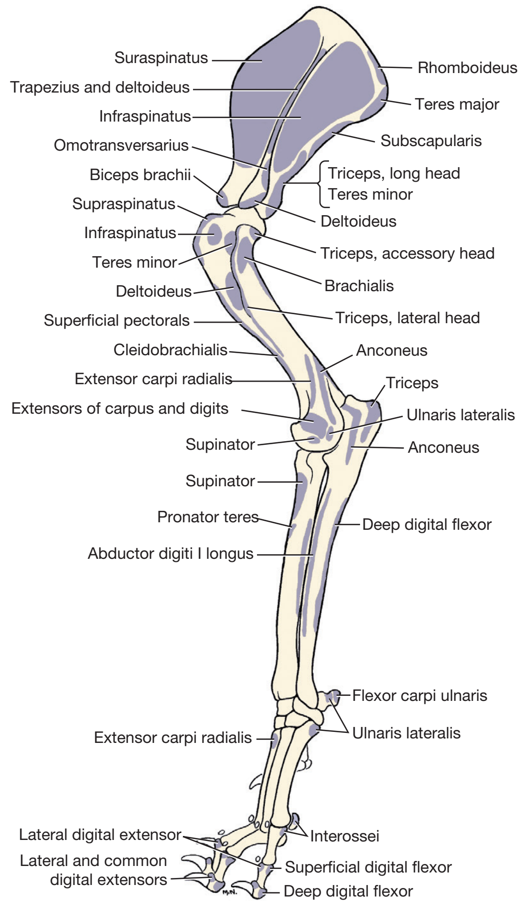


**Figure 5-20** Muscle attachment sites on the lateral forelimb skeleton. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

canine head and neck function mechanically like a cantilever beam. The weight of the head at the end of the "beam" produces a large neck flexion moment, which must be balanced by spinal ligaments and neck extensor muscles.

Dogs have a large muscle mass in shoulder adductors and elbow extensors because of the weight-bearing function of the forelimb. These antigravity muscles support the dog in weight bearing. Antigravity muscle groups for the forelimb include the following (see Figures 5-15 to 5-21):

- Shoulder extensors in closed chain and end-range open chain, and shoulder flexors in open chain from normal stance posture
- Shoulder adductors and abductors, depending on the weight-bearing posture
- Elbow extensors in closed chain, and flexors in open chain from stance
- Carpal and digit flexors in closed chain and end-range open chain, and extensors in open chain from normal stance

Prime movers for forelimb muscle groups are as follows:

- Shoulder flexors—latissimus dorsi, deltoideus, long head of the triceps brachii, teres major, teres minor
- Shoulder extensors—cleidobrachialis, biceps brachii, supraspinatus, brachiocephalicus
- Elbow flexors—biceps brachii and brachialis
- Elbow extensors—triceps brachii
- Carpal flexors—flexor carpi radialis, ulnaris lateralis, flexor carpi ulnaris
- Carpal extensors—extensor carpi radialis; carpal flexor and extensor muscles have flexor and extensor retinacula to keep tendons to the digits in proper alignment
- Digit flexors—superficial and deep digital flexors
- Digit extensors—common and lateral digital extensor muscles

Web Appendix 1 contains information on muscles, attachment sites, joint actions, analogous human muscles (muscle that performs a similar function in humans), and

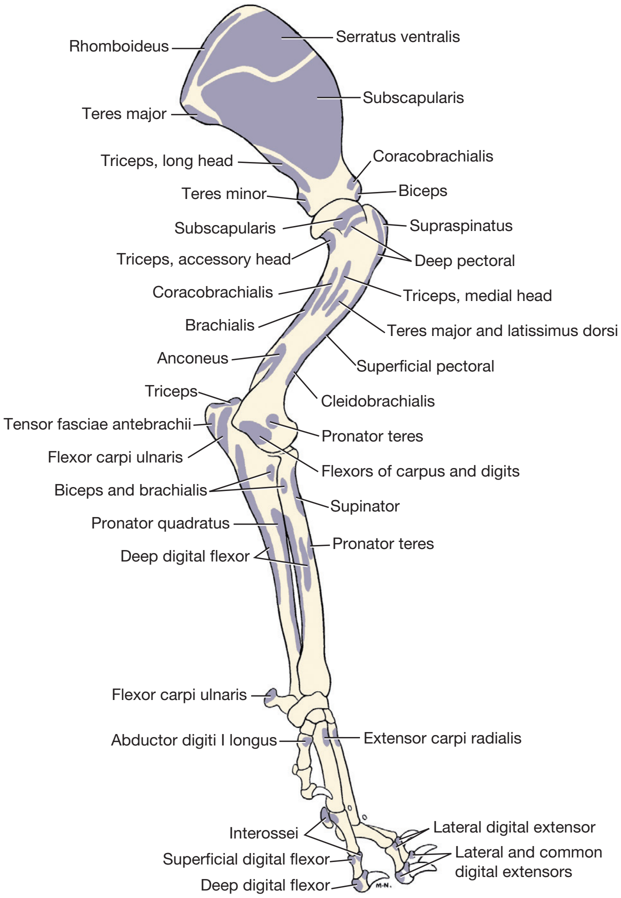


**Figure 5-21** Muscle attachment sites on the medial forelimb skeleton. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

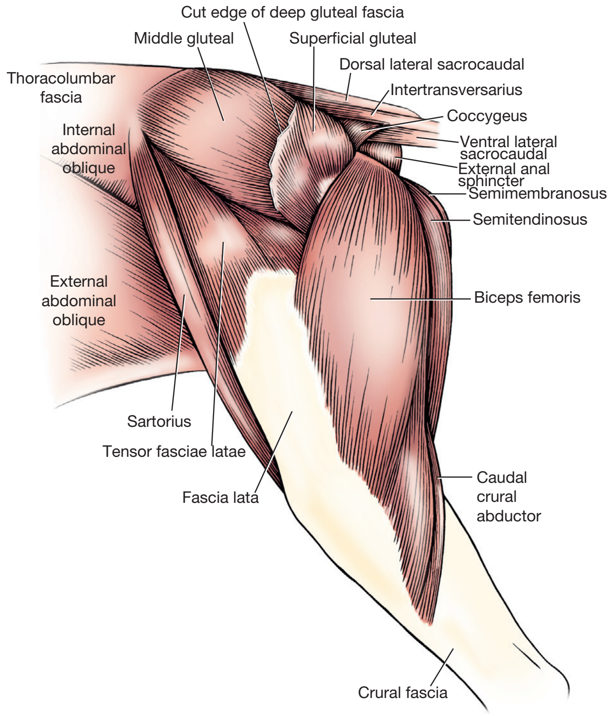


**Figure 5-22** Superficial muscles of the left lateral hindlimb. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

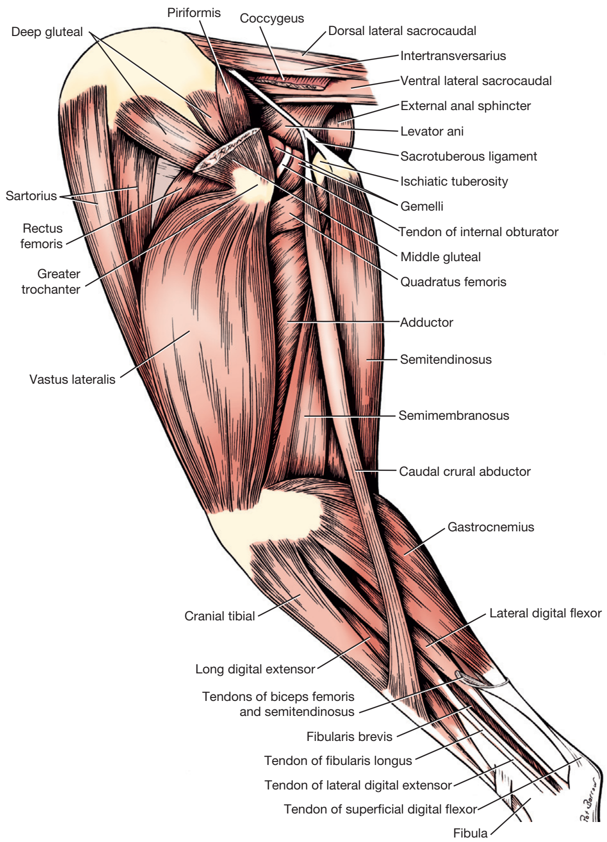


**Figure 5-23** Deep muscles of the left lateral hindlimb. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

innervation for prime movers of joint actions of the forelimb, hindlimb, and trunk and neck.

## Hindlimb

The antigravity muscles of the hindlimb support the dog in weight bearing or closed chain (Figures 5-22 to 5-30). In normal stance, the angle of the hip and stifle are both approximately 110 to 150 degrees, with perhaps more flexion in the stifle joint. This posture creates a constant requirement for large hip extensor and stifle extensor moments. The main antigravity muscle groups for closed-chain function in the hindlimb are as follows:

- Hip extensors in the sagittal plane
- Hip adductors in the frontal plane
- Stifle extensors
- Talocrural extensors
- Digit flexors

Prime movers for hindlimb muscles include the following:

- Hip flexors—iliopsoas, sartorius, tensor fasciae latae
- Hip extensors—gluteal muscles, biceps femoris, semitendinosus, semimembranosus
- Hip abductors—middle gluteal

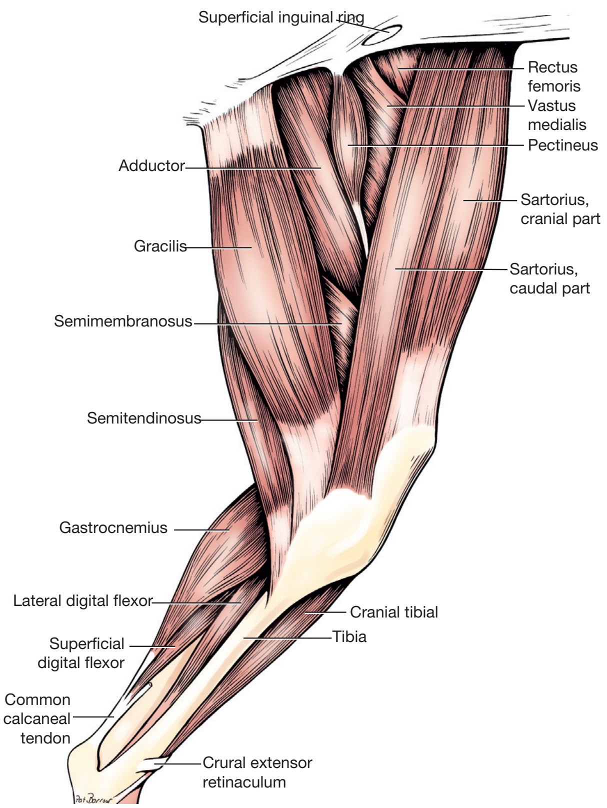


**Figure 5-24** Superficial muscles of the left medial hindlimb. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

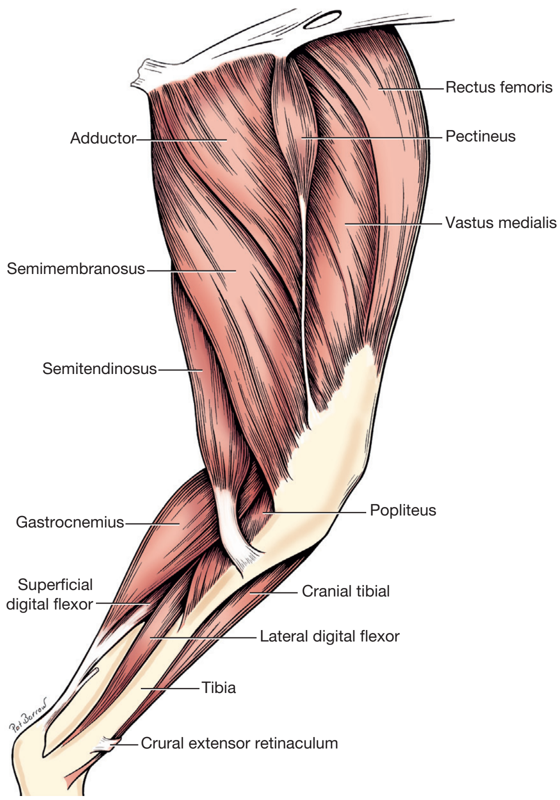


**Figure 5-25** Deep muscles of the left medial hindlimb. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

- Hip adductors—adductor magnus et brevis, pectineus
- Hip lateral rotators—internal and external obturator, gemelli
- Hip medial rotators—deep gluteal, semitendinosus
- Stifle extensors—quadriceps femoris: rectus femoris, vastus lateralis, vastus medialis, vastus intermedius
- Stifle flexors—biceps femoris, semitendinosus, semimembranosus
- Talocrural flexors—cranial tibial
- Talocrural extensors—gastrocnemius, superficial digital flexor
- Digit flexors—superficial and deep flexor
- Digit extensors—long digital extensor

Whereas the front limbs bear the majority of the static weight of the dog, the rear limbs provide the necessary propulsion for dynamic activity such as running and jumping.

As a result, the hamstring muscles are large and therefore have a large force capacity. The hamstring muscles in the dog are divided into cranial and caudal groups. The cranial group consists of the cranial parts of the biceps femoris and semimembranosus. The caudal group consists of the caudal parts of the biceps femoris and semimembranosus and the whole semitendinosus. The division into groups is based on electromyographic and functional characteristics. The cranial group functions primarily as antigravity muscles in which contraction is prominent at the point of foot placement and during the stance phase executing extension of the hip, whereas the caudal group contracts during the swing phase executing flexion of the stifle.[16] When compared with the human, the dog's ischial tuberosity is farther from the hip joint. Therefore the origin of the hamstrings on the ischial tuberosity provides a longer moment arm and therefore a relatively larger extensor moment about the hip joint. At the region of the femorotibial joint, the entire biceps femoris muscle courses superficial to the vastus lateralis muscle and, as an aponeurosis, joins the fascia lata and the fascia cruris. Based on these attachments, the biceps femoris is very effective in producing hip extension and stifle flexion, making this muscle an effective source of propulsion. The middle gluteal and adductor muscles are large, thick muscles. The superficial gluteal is small and located more on the lateral hip than the caudal hip.

The quadriceps femoris muscles are large. This is consistent with the constant and large stifle extension moment caused by the stifle position in normal stance.

At the talocrural joint, the canine gastrocnemius muscle is relatively large, and usually there is no soleus muscle. The long digital extensor muscle crosses the stifle and tarsal joints on the cranial surfaces. The tendons of the lateral digital extensor and peroneus brevis muscles travel

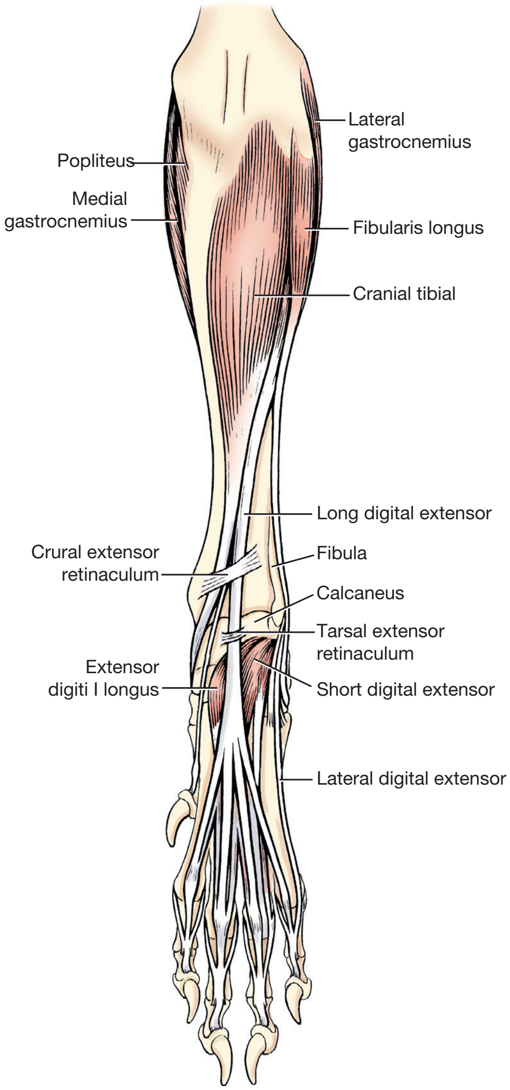


**Figure 5-26** Muscles of the left leg and hindpaw, cranial view. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

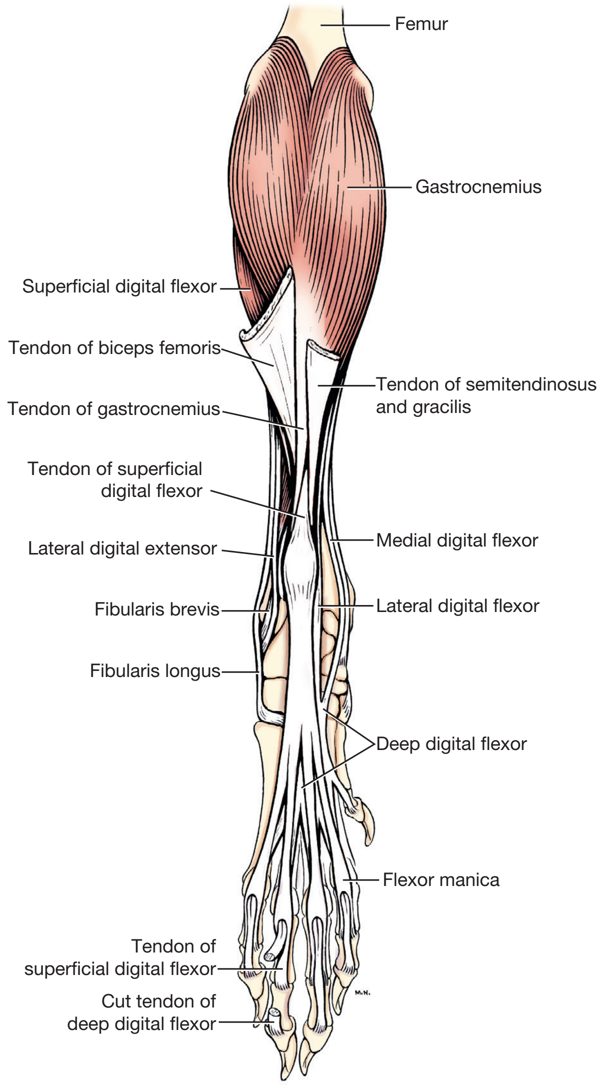


**Figure 5-27** Muscles of the left leg and hindpaw, caudal view. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

in a distinctive groove, the sulcus malleolaris lateralis, on the lateral malleolus.

## Trunk and Neck

Figures 5-31 to 5-36 display visual information about trunk and neck musculature. Canine muscles of the trunk and neck are characterized by nerve supply and embryologic development into hypaxial and epaxial muscles. The epaxial muscles develop locally in the trunk and are innervated by the dorsal rami of the spinal nerves. They include muscles that lie dorsal to the transverse processes of the vertebrae and have a main action of extension of the spine. The hypaxial muscles lie ventral to the transverse processes, are innervated by the ventral rami, and have a main action of flexion of the spine. The hypaxial group also includes the muscles of the abdominal and thoracic walls. The superficial dorsal trunk muscles act on the forelimb as well as the cervical spine and are innervated by the ventral rami.

Antigravity muscle groups for the spine are the trunk and neck extensors. Prime-mover muscles for spine motion include the following:
- Cervical spine flexion of C1-C2—longus capitis and longus colli
- Cervical spine flexion caudal to C2—sternomastoideus, longus capitis, and longus colli
- Cervical spine extension—splenius
- Cervical spine side bend—scalenus
- Rotation of the cervical spine—splenius and sternomastoideus
- Flexion of the trunk (thoracic and lumbar spine)—rectus abdominis muscle

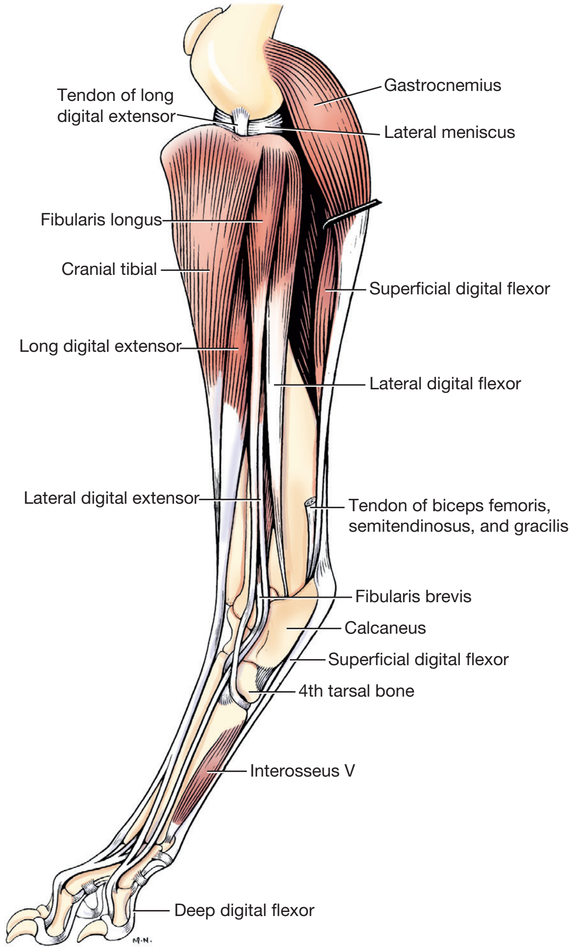


**Figure 5-28** Muscles of the left leg and hindpaw, lateral view. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

- Extension of the trunk—sacrospinalis or erector spinae muscles
- Side bend of the trunk—pars lumbalis of the external abdominal oblique
- Rotation of the trunk—longissimus and pars costalis of the external abdominal oblique

The posturing of the dog's head and neck as a cantilever beam produces a large amount of extensor or resisting neck flexion moment on the cervical spine. This posture creates a constant requirement for large neck extensor muscle activity. In response, there is a large amount of neck extensor muscle mass. Trunk flexor and extensor muscle masses are both large, reflecting a large requirement for trunk extensor and flexor muscle activity. The abdominal muscles have a big role in supporting the weight of the internal organs.

The deep dorsal trunk musculature, the extensor muscle groups, consists of three longitudinal muscle masses, each having many overlapping fascicles. The three masses are the iliocostalis muscle system laterally, the longissimus muscle system intermediately, and the transversospinalis muscle system medially. Different arrangements of fused portions of these three primary segmental muscle masses exist, based on the running abilities of the breed. Dogs that run fast may have longer fascicles, which allow muscles to attach to the spine for increased leverage. These breeds have fused caudal portions of the iliocostalis muscle with the longissimus muscle to form the erector spinae or sacrospinalis muscle group.

## Bursae

A synovial bursa (*bursae synoviales*) is a walled space containing a viscid fluid. The wall is composed of an outer tough layer, the membrana fibrosa, and a inner delicate layer, the membrana synoviales. Synovial bursae provide a gliding surface and a cushion for soft tissues as they pass over skeletal prominences and are found over bone and under muscle, tendon, ligament, fascia, or skin. Not all bursae are consistent structures, as some are present at birth (congenital) whereas others form later (acquired) in response to mechanical stresses.[17] Bursae can also be eliminated by aggressive inflammation as a result of infection or traumatic injury.

In the front limb of the dog, there is a bursa between the tendon of insertion of the infraspinatus muscle and the greater tubercle of the humerus. Frequently a second more proximal bursa can be found in association with the infraspinatus insertion.[17] Ossification of this bursa and the tendon of the infraspinatus tendon has been reported as a cause of lameness.[18] Diagnosis is suspected when pain is elicited by digital pressure over the site of the bursa and the tendon.

A bursa is present under the origin of the deep digital flexor muscle at the medial epicondyle of the humerus. The distal portion of the muscle is surrounded by a sheath extending from the level of the distal radius to the metacarpus. In some cases the sheath is less well defined with weaker walls and no fluid.[4]

Bursae are commonly associated with the extensor carpi radialis muscle. The joint capsule of the elbow forms a pocketlike structure under the proximal portion of the extensor carpi radialis muscle. Distally the extensor carpi radialis gives rise to two tendons, a medial tendon attaching to metacarpal II and a larger lateral tendon attaching to metacarpal III. In approximately half of all dogs there is a bursa that completely envelops both tendons as a complete fluid-filled tendon sheath. A separate bursa may be present under both tendons or just the lateral tendon at the level of the proximal row of carpal bones. A second bursa can be found under the lateral tendon at the level of the distal row of carpal bones. In other dogs fluid-filled sheaths or sacs are absent with a loosely meshed connective tissue in place.[4]

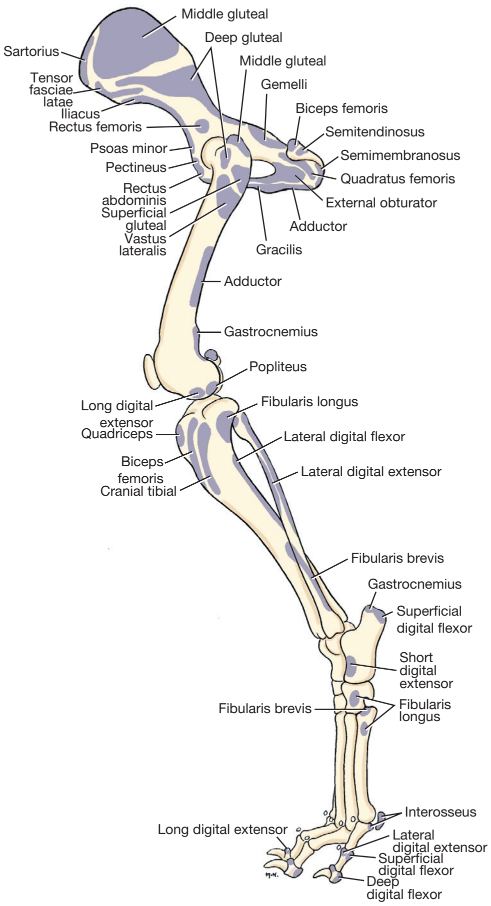


**Figure 5-29** Muscle attachment sites on the lateral hindlimb skeleton. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

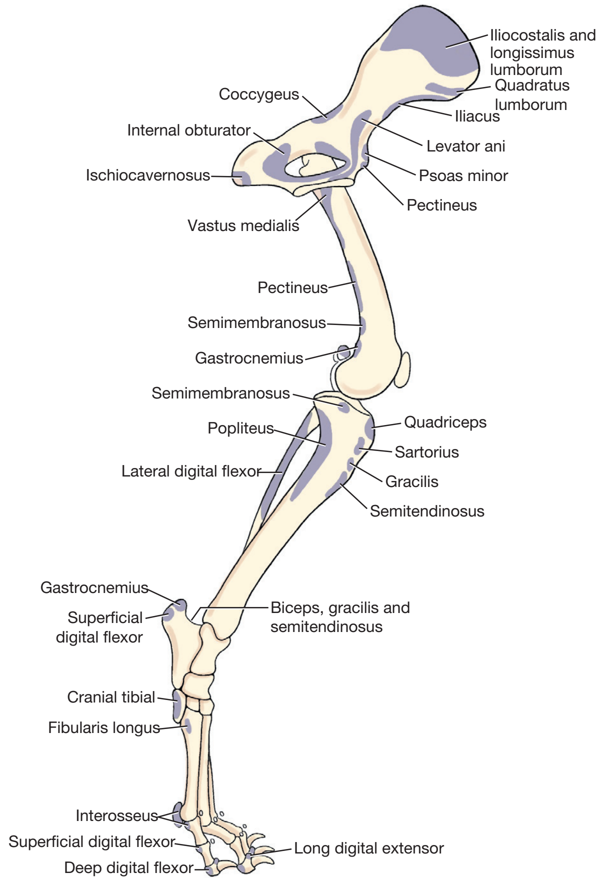


**Figure 5-30** Muscle attachment sites on the medial hindlimb skeleton. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

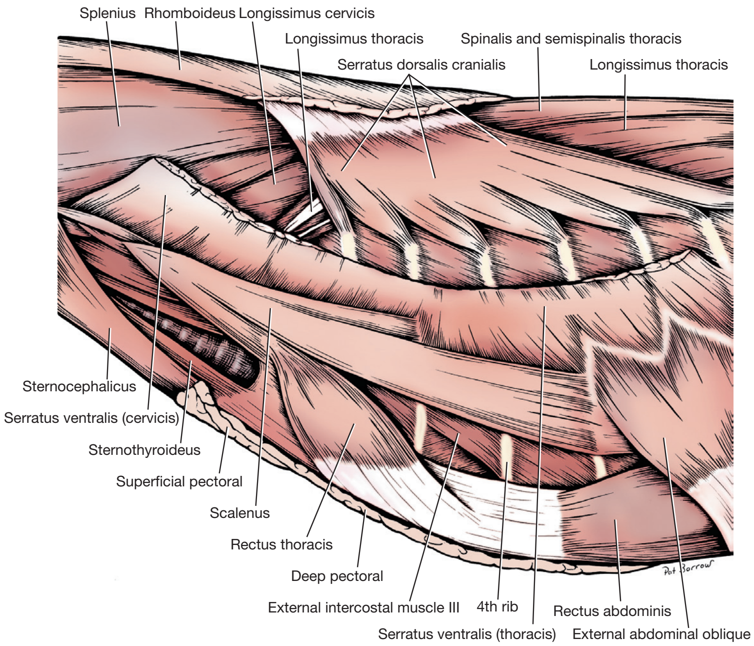


**Figure 5-31** Muscles of the left lateral neck and thorax. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

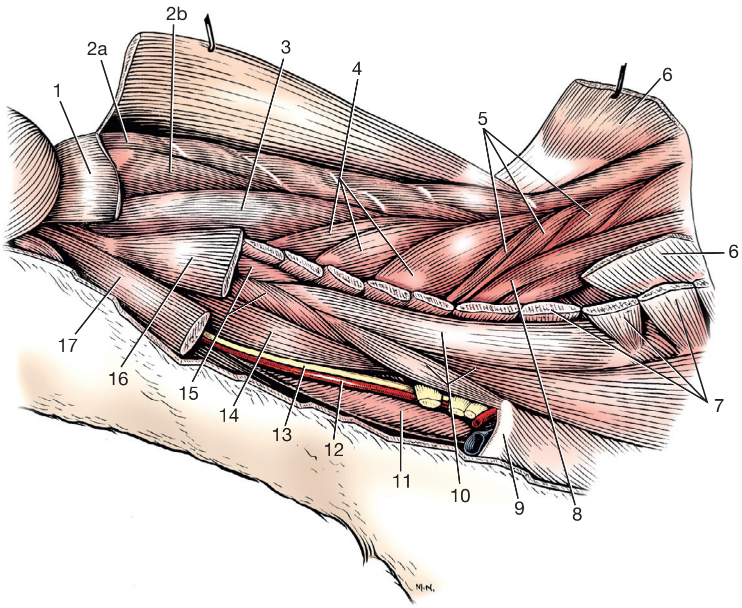


**Figure 5-32** Deep muscles of the left lateral neck. (From Evans HE, de Lahunta A: *Miller's guide to the dissection of the dog,* ed 7, Philadelphia, 2010, WB Saunders.)

1. Splenius
2. Semispinalis capitis:
   a. Biventer cervicis
   b. Complexus
3. Longissimus capitis
4. Longissimus cervicis
5. Longissimus thoracis
6. Serratus dorsalis cranialis
7. Serratus ventralis
8. Iliocostalis thoracis
9. First rib
10. Scalenus
11. Esophagus
12. Common carotid artery
13. Vagosympathetic trunk
14. Longus capitis
15. Intertransversarius
16. Omotransversarius
17. Mastoid part of cleidocephalicus

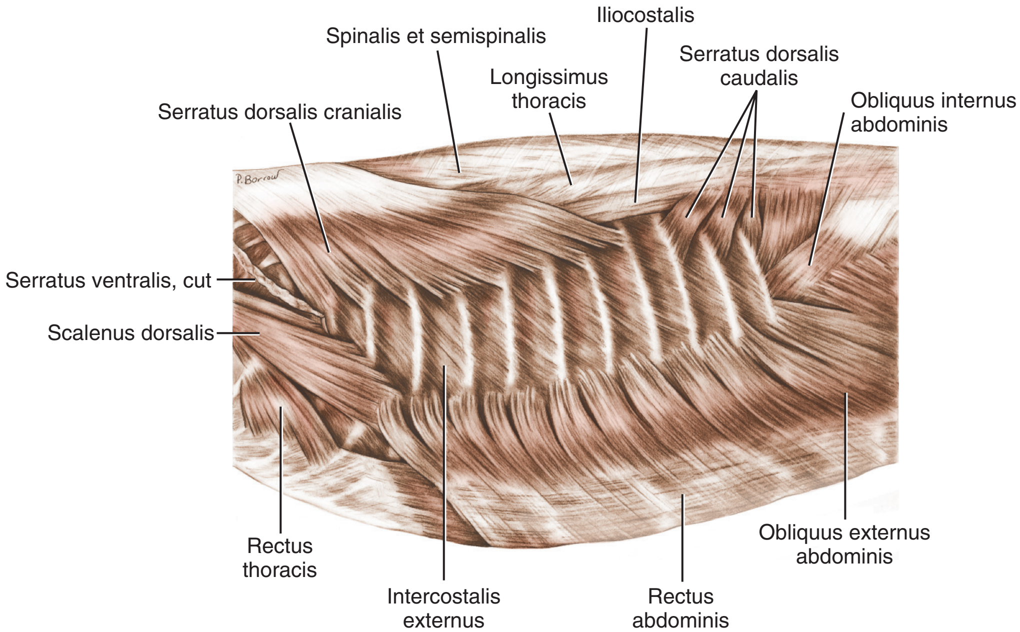


**Figure 5-33** Superficial muscles of the left lateral thoracic cage. (From Evans HE: *Miller's anatomy of the dog,* ed 4, Philadelphia, 2013, WB Saunders.)